How Much Can LLMs Hallucinate? An Upper Bound via Coupling and Ambiguity

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Abstract

The dominant lineage in hallucination theory bounds *frequency* from below — calibrated LLMs *must* hallucinate at some rate. We bound *size* from above: the goal-conditional displacement of the post-update model state from the goal-marginal, under coupled belief-goal inference. The two axes are orthogonal — how often versus how much an LLM hallucinates. The size axis was open: posterior-stability machinery from Bayesian inverse problems had not been brought to coupled inference.

The main theorem bounds the displacement by transferred information,

$$\mathbb{E}\,\|\Delta M_{\text{bias}}\| \le C\sqrt{I(G; M_{\tau^+} \mid e, M_{\tau^-})},$$

under standard regularity conditions and *independent of any architectural-class commitment*. Under a named coupling-attenuation hypothesis, this factors as $C\sqrt{\kappa_{\text{processing}} \cdot I(G; \Omega_\tau \mid e_\tau, M_{\tau^-})}$ — coupling strength times residual ambiguity. Two routes deliver C. A transport-inequality track uses a Talagrand T_2 inequality on the post-update law, verifiable via log-Sobolev concentration or dimension-free sub-Gaussian moments. A Fisher-Rao geometry track yields *two* universal dimension-free constants on d_{FR}: $C = \sqrt{2}$ locally tight under parameterization-invariance + Riemannian structure + KL-coordinate matching + uniform-locality, and $C = \pi/\sqrt{2}$ globally valid under parameterization-invariance alone (with the $\pi/2$ overhead being exactly the worst-case arc-chord ratio at the unit-sphere antipode). The (PI) commitment is load-bearing: no chart-independent universal constant exists under Euclidean coordinates (a two-chart scale-family construction), and the four-condition triple uniquely pins Fisher-Rao + $\sqrt{2}$ as the locally-tight constant. A direct Hellinger statement gives a third universal constant $1/\sqrt{2}$ on the chord — companion to the global Fisher-Rao bound on the same unit L^2-sphere.

Architectures partition by goal-update coupling into a monotonic ladder — Separated, Partial, Coupled. A structural lemma derives Coupled status for transformer attention from connectivity alone (under non-degenerate attention). For conjugate-Gaussian inverse problems, the bound's Euclidean translation carries prefactor $L_{\text{post}}\sigma$, bounded by $\tau/2$ uniformly and vanishing in extreme observation-noise limits.

1 Introduction

Recent hallucination theory bounds the *frequency* with which calibrated language models must produce false outputs as a consequence of training-set structure or calibration constraints —

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Kalai and Vempala 2023 [1] on singleton-fact rates, Kalai et al. 2026 [2] on training-and-evaluation-incentive origins, Karbasi et al. 2025 [3] on detection impossibility, Wu et al. 2024 [4] on VC-dimension obstructions, Suzuki et al. 2025 [5] on probabilistic negligibility. Hallucination in this lineage is treated as architecture-agnostic.

Bayesian inverse-problems theory in the lineage of Stuart 2010 [6] has spent fifteen years building Lipschitz-stability bounds on the posterior under perturbations of the prior, the likelihood, and the data — Sprungk 2020 [7], Cvetković and Lie 2025 [8], Dolera and Mainini 2023 [9], Garbuno-Iñigo et al. 2023 [10], Hosseini et al. 2025 [11]. The transport-inequality apparatus driving these bounds — Otto-Villani, Bakry-Émery, Gozlan, Bobkov-Götze — is mature and verified across decades of inverse-problems practice.

Neither lineage engages the implicit belief-goal-coupling structure of attention-based architectures — a structure intrinsic to LLMs by construction (Lemma 3.5), not an incidental property of training.

Engaging this structure directly reframes the question: how much can an LLM be pulled away from what its evidence supports by the goal it is asked to satisfy? This paper bounds the displacement.

This paper composes the two Apply the chain rule of relative entropy directly to the *post-update* law, marginalized over the goal: $\mathbb{E}_G\left[\mathrm{KL}(P_{M_{\tau^+}|e,M_{\tau^-},G} \,\|\, P_{M_{\tau^+}|e,M_{\tau^-}})\right] = I(G; M_{\tau^+}|e_\tau, M_{\tau^-})$. The right-hand side is the *transferred* goal-information that actually enters the post-update model state. Combined with a Fisher-Rao second-order expansion on the post-update statistical manifold, this delivers an upper bound on goal-conditional displacement:

$$\mathbb{E}\,\|\Delta M_{\text{bias}}\| \;\le\; C \cdot \sqrt{I(G; M_{\tau^+} \mid e_\tau, M_{\tau^-})}\,. \tag{1}$$

Under a parameterization-invariance commitment at full Markov-morphism strength — plus Riemannian structure and KL-coordinate matching — Čencov's 1982 uniqueness theorem forces Fisher-Rao with constant $C = \sqrt{2}$. *Universal, dimension-free, no domain-specific parameters.* A scale-family no-go theorem rules out any coordinate-independent universal constant for Euclidean chart norms — making the (PI) commitment load-bearing, not coincidental. We call this *Track 2*.

A parallel *Track 1* derivation via a Talagrand transport inequality on the same post-update law recovers the canonical Stuart-school cascade constant $C_{T_2} = 2L^2_{\text{post}}/\rho_{\text{LSI}}$ as a special case, with $C = \sqrt{C_{T_2}}$ on W_2. Track 2's universal constant is what makes the composition novel; Track 1 demonstrates that the composition contains the existing Bayesian-inverse-problems lineage as a strict sub-case under standard log-Sobolev concentration. Both flow from the same chain-rule move on the post-update law.

The architectural reading Under a coupling-attenuation hypothesis (H_κ), the bound factors:

$$\mathbb{E}\,\|\Delta M_{\text{bias}}\| \;\le\; C \cdot \sqrt{\kappa_{\text{processing}} \cdot I(G;\, \Omega_\tau \mid e_\tau, M_{\tau^-})}\,, \tag{2}$$

into an *architectural coupling factor* $\kappa_{\text{processing}}$ (how much of the available goal-information the architecture's update mechanism transfers) and a *residual ambiguity factor* $I(G; \Omega \mid e, M)$ (how much the goal could in principle resolve about the latent world-state beyond what the evidence already pins down). Architectures partition by their goal-update topology into a monotonic ladder — Class 1 (Separated), Class 2 (Partial), Class 3 (Coupled). For Class 1, $\kappa_{\text{processing}} \le 1$ holds automatically by data-processing on the Markov chain $G \to \Omega \to M_{\tau^+}$. For Class 2 and Class 3, (H_κ) is a named structural commitment with documented sufficient conditions and a worst-case witness (parrot architecture, $\kappa_{\text{processing}} = \infty$) excluded by scope.

Plain decoder-only transformer attention is *Class 3 (Coupled) by construction* in a precise structural sense: every internal computation has a directed-graph path from any input position whenever attention weights are non-degenerate (Lemma 3.5, by induction on layer depth, robust to RMSNorm / FlashAttention / causal masking / sliding-window). The structural Coupled-class claim does not depend on the token-distribution-as-belief-state idealization that the bias-bound *application* additionally invokes via the in-context-learning correspondence [12–15].

Contributions

1. **An architectural classification.** Goal/Update Coupling Class — Class 1 (Separated), Class 2 (Partial), Class 3 (Coupled) — partitions architectures monotonically by goal-update topology, grounded in the Pearl-blanket reading[16] of the Markov-blanket apparatus. Lemma 3.5 derives Class 3 status for transformer attention from connectivity alone.
2. **An umbrella upper bound** (1) on goal-conditional displacement of the post-update model state, derived via chain rule on the post-update law. Two named tracks supply C — a transport-inequality cascade (Track 1) and a Fisher-Rao geometry route (Track 2).
3. **A no-go that forces a coordinate commitment.** No coordinate-independent universal constant exists for Euclidean chart norms (a scale-family construction). Under the (PI) + (R) + (K) triple, Čencov's uniqueness theorem forces Fisher-Rao + $\sqrt{2}$ universally — no further freedom.
4. **An architectural-corollary form** (2) under (H_κ), recovering the $\kappa \times \mathcal{A}$ factorization that gives the bound its operational reading.

Section 3 sets up the architectures and the (PI), (R), (K) axioms. Section 4 states the umbrella theorem with both tracks, the no-go, and the architectural corollary. Section 5 sketches the mechanism — both proofs flow from the post-update chain rule. Limitations and the bound's relation to the frequency-lower-bound lineage are in Section 6.

2 Related Work

Two literatures sit on either side of the gap this paper closes; brief categorization (full positioning across all six adjacent strands in Appendix F).

LLM hallucination theory has produced architecturally-agnostic frequency lower bounds on calibrated language models — calibration-constrained singleton-fact rates, training-and-evaluation-incentive origins, learnability and computability obstructions, probabilistic-negligibility positive results (cited in §1). These results derive from training-set structure or capacity considerations and do not engage the architecture's belief-goal-coupling structure. They bound *how often* a calibrated model must hallucinate.

Bayesian inverse-problems posterior stability (the Stuart school) has produced mature Lipschitz-continuity bounds on the posterior under perturbations of the prior, the likelihood, and the data, using transport-inequality machinery (Otto-Villani, Bakry-Émery, Gozlan, Bobkov-Götze) and Fisher-information-flavored constants in optimal-transport frameworks (cited in §1). These results address well-posedness of Bayesian inference; they have not been applied to belief-goal-coupled inference. The machinery bounds *how stable* a Bayesian posterior is to specified perturbations.

The two bound different things, sit in different research traditions, and do not engage with each other. The chain-rule of relative entropy applied directly to the post-update law (Lemma 5.1) is the bridging move: it converts the goal-coupling problem into a transferred-information quantity, which the Stuart-school transport machinery and the Fisher-Rao expansion both bound at the post-update law. *Both* lineages enter — Track 1 inherits the Otto-Villani / Bakry-Émery cascade; Track 2 deploys Čencov uniqueness on the manifold the Stuart school operates on. The composition is the contribution; the architectural reading via Class 1 / Class 2 / Class 3 makes the bound operational.

Adjacent strands — Fisher-Rao geometry for Bayesian sensitivity analysis, Owhadi-Scovel-Sullivan brittleness no-gos, mean-field variational inference stability, the Markov-blanket apparatus from active inference — sit close enough to require explicit cite-and-distinguish but do not resolve the gap. Detailed positioning across all strands is in Appendix F.

3 Setup

We work with goal-conditioned probabilistic models in a Bayesian-update setting. The state at time τ^- before update consists of an epistemic component M_{τ^-} (the model state, an element of a parameter manifold $\mathcal{M}$) and a goal component G. An event e_τ arrives — a token, an observation, a query response — and the model updates to M_{τ^+}. The latent world-state Ω_τ is what the event is evidence about. The architecture's update mechanism produces a post-update model state M_{τ^+} whose law depends — possibly stochastically — on the goal G. The goal-conditional post-update law is $P_{M_{\tau^+}|e_\tau,M_{\tau^-},G}$; the goal-marginal post-update law $P_{M_{\tau^+}|e_\tau,M_{\tau^-}}$ is the natural baseline against which goal-coupling is measured (the operating-distribution average over goals).

3.1 The bias quantity

The *goal-conditional bias* measures the displacement of the goal-conditional law from the goal-marginal:

$$\|\Delta M_{\text{bias}}(G)\| \;:=\; \|P_{M_{\tau^+}|e_\tau,M_{\tau^-},G},\; P_{M_{\tau^+}|e_\tau,M_{\tau^-}}\|_{\mathcal{M}}, \tag{3}$$

a random variable in G. Throughout, $\|\cdot\|_{\mathcal{M}}$ denotes either the Wasserstein distance W_2 on the induced model-distribution (Track 1) or the Fisher-Rao geodesic distance d_{FR} on the statistical-manifold sub-case (Track 2). Euclidean-on-parameters is the choice Section 4.1 rules out.

3.2 Goal/Update Coupling Class

Architectures partition by whether G has a causal path into the belief-update computation. The partition is monotonic in coupling strength.

Table 1: Goal/Update Coupling Class — partition by conditional-independence structure.

Class	Goal/update coupling	Examples
Class 1 (Separated)	Holds by construction — estimator has no causal path from G	Kalman filter + LQR; modular RL with separate world model
Class 2 (Partial)	Holds for some pathways, fails for others	Biological cortex; hybrid systems with separate preprocessing
Class 3 (Coupled)	Fails by construction — G is causally upstream of every computation	Transformer LLM (attention processes goals and observations together — Lemma 3.5)

Class 1 (Separated) is the architectural-separation case studied throughout Bayesian inverse-problems theory. Class 3 (Coupled) is the case the existing hallucination-theory lineage doesn't engage geometrically. Class 2 (Partial) is genuinely intermediate — hybrid LLM systems with retrieval, tools, scratchpads, or external memory — and lives at the system level rather than the architecture level. The Pearl-blanket reading[16] of the Markov-blanket apparatus grounds the partition; we do not adopt the Friston-blanket metaphysical reading.

The bound's architectural-corollary form ((2)) factors transferred goal-information through an *attenuation ratio* — the proportion of in-principle-resolvable goal-information that the architecture's update mechanism actually transfers into the post-update model state:

$$\kappa_{\text{processing}} \;:=\; \frac{I(G;\, M_{\tau^+} \mid e_\tau,\, M_{\tau^-})}{I(G;\, \Omega_\tau \mid e_\tau,\, M_{\tau^-})}. \tag{4}$$

Conditioning on M_{τ^-} matters: prior correlation between goals and prior model state — present even in Class 1 — would inflate the measure otherwise. For Class 1, the data-processing inequality on the chain $G \to \Omega_\tau \to M_{\tau^+}$ (Markov conditional on (e_τ, M_{τ^-})) gives $\kappa_{\text{processing}} \leq 1$ automatically. For Class 2 and Class 3, $\kappa_{\text{processing}} \leq 1$ is a named structural commitment; the parrot architecture $M_{\tau^+} := G$ with $G \perp\!\!\!\perp \Omega_\tau$ given (e_τ, M_{τ^-}) produces $\kappa_{\text{processing}} = \infty$, explicitly outside scope.

3.3 Axioms

The universal-constant route (Section 5.2, Track 2) commits to three axioms on the bound's metric $d_{\mathcal{M}}$:

Hypothesis 3.1 ((PI) Sufficient-statistic invariance)**.** *The bias bound's metric on $\mathcal{M}$ is invariant under sufficient statistics — equivalently, under Markov morphisms / congruent embeddings of statistical manifolds in the sense of Čencov 1982[17].*

(PI) at full Markov-morphism strength is what Čencov's uniqueness theorem actually requires. Smooth-coordinate invariance alone admits an infinite family of metrics; the strictly-stronger Markov-morphism class singles out Fisher-Rao up to global scalar.

Hypothesis 3.2 ((R) Riemannian structure)**.** *The bound's metric $d_{\mathcal{M}}$ derives from a smooth Riemannian metric on $\mathcal{M}$.*

(R) filters out divergence forms (TV, χ^2, Hellinger-as-divergence) that satisfy (PI) as f-divergences but are not metric distances on $\mathcal{M}$ in the Riemannian sense.

Hypothesis 3.3 ((K) KL second-order matching)**.** $\mathrm{KL}(P_\theta \,\|\, P_{\theta+\delta}) = \frac{1}{2} d^2_{\mathcal{M}}(P_\theta, P_{\theta+\delta}) + O(d^3_{\mathcal{M}})$.

(K) fixes the global scalar Čencov leaves free, pinning the bound's constant C at $\sqrt{2}$. (R) and (K) are both implicit in any "$\mathbb{E}\, d \leq C\sqrt{I}$" theorem-shape with second-order matching; we name them explicitly so the uniqueness statement of Theorem 4.5 has its hypotheses fully stated.

The transport-inequality route (Section 5.1, Track 1) and the dimension-restricted slice-wise refinements (Theorem 4.1's Track 2 locally-tight instantiation) introduce additional regularity hypotheses inline with the theorems that invoke them. Throughout the paper we restrict to the *statistical-manifold sub-case*:

Hypothesis 3.4 ((H1) Statistical-manifold sub-case)**.** *The model state $M_t \in \mathcal{M}$ corresponds to a probability distribution P_{M_t} over latent world-states, and $\mathcal{M}$ is locally a statistical manifold in the sense of Amari and Nagaoka 2000[18]; $\mathcal{M}$, the goal-space, and the event-space are* standard Borel*, so regular conditional probabilities exist as Markov kernels[19] Theorem 6.3.*

(H1) is satisfied by parametric Bayesian posterior families, exponential families, conjugate priors, and the parametric-belief-state subclass of LLMs that admits a natural distributional reading. The standard-Borel regularity is what makes the post-update chain rule (used in both Tracks) hold in the abstract-spaces form[20] Theorem 3.4;[21] Theorem 5.4.

3.4 Coupled-class attention connectivity

The architectural classification's load-bearing application — that *plain decoder-only transformer attention is structurally Class 3 (Coupled)* — is a property of attention connectivity, not of any token-distribution-as-belief-state idealization.

Lemma 3.5 (Coupled-class attention connectivity)**.** *Let $\mathcal{G}_\theta$ be the directed computational graph of an autoregressive transformer with parameters θ, taking input sequence $X_{1:t-1}$ and producing internal activations $\{h^{(i)}_\ell\}_{\ell,i}$ across layers ℓ and positions i. If positions $i_G \subseteq \{1, \ldots, t-1\}$ contain goal/prompt tokens and positions i_E contain evidence tokens with $i_G \cap i_E = \emptyset$, then for every layer $\ell \geq 1$ and position $j \geq \max(i_G \cup i_E)$, the activation $h^{(j)}_\ell$ has a directed-graph path from at least one i_G position whenever attention weights are non-degenerate on the i_G row.*

The proof is a one-induction over layer depth, robust to RMSNorm / GroupNorm / FlashAttention (position-wise / mathematically-equivalent), causal masking (preserves $\mathrm{Attn}^{(j,i_G)}_\ell \neq 0$ for $i_G \leq j$), and sliding-window or sparse attention (preserves connectivity by composition across layers when the window pattern admits a multi-hop path). Architectures with explicit goal-to-output attention masking, or token-dropping at intermediate layers, fall outside the lemma's scope and are appropriately Class 2 (Partial). The full proof is in Appendix E.5.

Lemma 3.5 establishes a *downstream-output graph-reachability* claim: at every layer and every position causally downstream of the goal and evidence, the activation has a directed-graph path back to at least one goal position. The bias-bound *interpretation* additionally invokes the in-context-learning correspondence[12–15] between the next-token distribution and an implicit posterior; the structural Coupled-class claim of Lemma 3.5 is robust without it.

4 Main Results

The umbrella theorem bounds the goal-conditional displacement of the post-update model state by transferred goal-information. Two named tracks supply the constant — a transport-inequality cascade (Track 1) and a Fisher-Rao geometry route (Track 2) — both flowing from a common chain-rule move on the post-update law (mechanism in Section 5).

Theorem 4.1 (Umbrella bound on goal-conditional displacement)**.** *Under (H1) and the regularity hypotheses below, for fixed event-prior pair (e_τ, M_{τ^-}) with $\mathbb{E}$ taken over $G \sim P(G \mid e_\tau, M_{\tau^-})$:*

$$\mathbb{E}\,\|\Delta M_{bias}\| \;\leq\; C \cdot \sqrt{I(G;\, M_{\tau^+} \mid e_\tau,\, M_{\tau^-})}, \tag{5}$$

with C supplied by either of two routes:

(Track 1, transport-inequality): *Under additional (H2′) — a Talagrand T_2 inequality on the post-update law — $C = \sqrt{C_{T_2}}$ with metric W_2. The constant C_{T_2} recovers the canonical Stuart-school cascade value $2L^2_{post}/\rho_{LSI}$ as a special case under standard log-Sobolev concentration.*

(Track 2, Fisher-Rao): *Under additional (H4′) — uniform-locality on the goal-conditional slice-wise KL — $C = \sqrt{2}$ with metric d_{FR}. The constant is universal, dimension-free, and carries no domain-specific parameters.*

Hypothesis 4.2 ((H2′) Talagrand T_2 on the post-update law)**.** *The post-update law $P_{M_{\tau^+}|e,M_{\tau^-}}$ satisfies $W_2^2(P,\, P_{M_{\tau^+}|e,M_{\tau^-}}) \leq C_{T_2} \cdot \mathrm{KL}(P \,\|\, P_{M_{\tau^+}|e,M_{\tau^-}})$ for all probability measures P. Sufficient conditions in Appendix B: log-Sobolev on the post-update density (Otto-Villani $C_{T_2} = 2/\rho_{LSI}$); dimension-free sub-Gaussian Lipschitz concentration*[22]*.*

Hypothesis 4.3 ((H4′) Uniform local regime)**.** $\operatorname{ess\,sup}_g \mathrm{KL}(P_{M_{\tau^+}|e,M_{\tau^-},G=g} \,\|\, P_{M_{\tau^+}|e,M_{\tau^-}}) \leq \delta_\star$ *for some $\delta_\star \in (0,1)$. Every goal-conditional slice — not merely the average — is uniformly close to the goal-marginal on the statistical manifold. (H4′) is strictly stronger than $\mathbb{E}_G[\mathrm{KL}_g] = I \ll 1$; small mean does not imply small ess-sup.*

Remarks on Theorem 4.1.

The two tracks bound different metrics with the same square-root-in-information shape. Track 1 lives on W_2, Track 2 on d_{FR}. Both deliver bounds of form $\mathbb{E}\, d \leq C\sqrt{I_M}$ where $I_M = I(G; M_{\tau^+} \mid e_\tau, M_{\tau^-})$ is transferred goal-information. The two are not "different scalings"; they are bounds on different metrics with the same information dependence.

The bound is unconditional in the architectural classification. Both Tracks bound transferred goal-information directly. (H1)–(H2′) and (H1)–(H4′)–(PI)–(R)–(K) are regularity hypotheses on the post-update law and the metric. Neither makes a structural commitment about how the goal G enters the architecture's update mechanism. The architectural reading enters at corollary level via (H_κ) below.

Track 2's constant is universal and dimension-free. Track 1's C_{T_2} is family-specific (depends on the post-update concentration). Track 2's $\sqrt{2}$ is the same constant on every statistical manifold under the (PI)+(R)+(K) triple, by Čencov uniqueness (Theorem 4.5 below).

Track 1 contains the Bayesian-inverse-problems lineage. In the conjugate-Gaussian Class 1 (Separated) case with invertible data channel, Track 1's C_{T_2} recovers $2L^2_{\mathrm{post}}/\rho_{\mathrm{LSI}}$ exactly — the canonical cascade constant from Stuart 2010 [6] and the lineage that follows. The composition contains the existing literature as a strict sub-case.

Track 1 outside (H2′). Heavy-tailed post-updates without log-Sobolev or dimension-free sub-Gaussian concentration fall outside Track 1's T_2 scope. A weaker T_1 form via bounded support gives the distance bound at W_1 rather than W_2 — adequate for the distance reading, weaker on the squared-distance side. Track 2 under (PI)+(R)+(K) does not require (H2′) and remains available.

4.1 A no-go that forces (PI)

The universal-constant route (Track 2's $\sqrt{2}$) is not coincidental. A scale-family construction shows that no coordinate-independent universal constant exists for Euclidean chart norms — making the (PI) commitment load-bearing.

Theorem 4.4 (No-go on Euclidean chart norms)**.** *Suppose there were a constant $C_0 < \infty$ — independent of any chart on any statistical manifold — such that for every chart ϕ on $\mathcal{M}$ and every goal-coupled architecture satisfying (H1)+(H2′) with non-zero post-update displacement, $\mathbb{E}\,\|\Delta M_{bias}\|_{\mathrm{Eucl},\phi} \leq C_0 \sqrt{I(G; M_{\tau^+} \mid e_\tau, M_{\tau^-})}$. Then no such C_0 exists.*

The proof is short: a chart rescaling $\phi \mapsto a\phi$ scales the chart-Euclidean Wasserstein distance linearly while leaving the right-hand side (chart-invariant information) unchanged; taking $a \to \infty$ contradicts the fixed $C_0\sqrt{I}$. Full statement and proof in Appendix E.3. The Gaussian scale family with chart σ versus $\log\sigma$ is the cleanest illustration.

Remark Theorem 4.4 is *adjacent in shape but opposite in direction* to Owhadi-Scovel-Sullivan's Bayesian-brittleness results[23,24]. Their no-go is for *arbitrary stability under finite information* in fixed metrics; ours is for *a universal constant absent a coordinate-invariance commitment*, in a setting where (H2′)'s post-update concentration explicitly excludes their brittleness regime. The two no-gos constrain different things; neither implies the other.

4.2 Čencov uniqueness and sharpness

Track 2's constant $\sqrt{2}$ is not just universal under (PI)+(R)+(K) — it is the *unique sharp* such constant.

Theorem 4.5 (Uniqueness and sharpness of the Fisher-Rao $+\sqrt{2}$ bound)**.** *Let $\mathcal{M}$ be a statistical-manifold sub-case (H1), and let the bound take the umbrella form $\mathbb{E}\, d_{\mathcal{M}}(\Delta M_{bias}) \leq C\sqrt{I(G; M_{\tau^+} \mid e_\tau, M_{\tau^-})}$ under the constraints (PI)+(R)+(K). Then:*

(a) *$d_{\mathcal{M}}$ is uniquely the Fisher-Rao geodesic distance.*

(b) *No constant $C < \sqrt{2}$ uniformly bounds $\mathbb{E}\, d_{FR}/\sqrt{I_M}$ across the (H4′) regime: there exist conjugate-Gaussian Class 1 instances on which $\mathbb{E}\, d_{FR}^2/I_M \to 2$ as the goal variation shrinks at fixed I_M. Hence $C = \sqrt{2}$ is the unique sharp upper-bound constant under (PI)+(R)+(K)+(H4′).*

Proof in Appendix E.4. Part (a) is Čencov's 1982 uniqueness theorem applied to $\mathcal{M}$: the Fisher information metric is the unique (up to global positive scalar) Riemannian metric invariant under Markov morphisms. Part (b) is direct verification on the conjugate-Gaussian saturation.

Remark Alternative chart-independent commitments (TV with Pinsker, Hellinger-as-divergence, χ^2, Rényi) are all defeated: TV-Pinsker is non-tight (the constant is universal but the inequality is loose except at coincident measures); Hellinger-as-divergence and χ^2-as-divergence violate (R); switching divergences on the right-hand side violates (K). Under the cascade structure (K) and the metric framing (R), the no-go (Theorem 4.4) tells us *exactly* what to commit to — (PI) — and the universal $\sqrt{2}$ falls out.

4.3 The architectural corollary

The umbrella theorem bounds displacement by *transferred* goal-information. The architectural reading factors this transferred information into a structural-architectural quantity (where can the goal flow?) and an information-theoretic quantity (how much could the goal in principle resolve?).

Hypothesis 4.6 ((H$_\kappa$) Bounded architectural attenuation)**.** *$\kappa_{processing} \leq \kappa^* < \infty$, with κ^* a named architectural constant (typically 1 for non-amplifying architectures).*

Corollary 4.7 (Architectural factorization)**.** *Under (H1) + the Track-specific hypotheses + (H$_\kappa$) at level κ^*, the bound factors as*

$$\mathbb{E}\,\|\Delta M_{bias}\| \;\leq\; C\sqrt{\kappa^* \cdot I(G;\, \Omega_\tau \mid e_\tau,\, M_{\tau^-})}, \tag{6}$$

with C supplied by Track 1 or Track 2 as in Theorem 4.1.

Remarks on Corollary 4.7.

The factorization is the contribution-headline. Coupling strength $\kappa_{\text{processing}}$ × residual ambiguity $I(G; \Omega \mid e, M)$ — architecture × information. The unconditional theorem Theorem 4.1 is the structural backbone whose math holds without (H$_\kappa$); the corollary is the operational reading.

Class 1 (Separated) gives (H$_\kappa$) automatically. The Markov chain $G \to \Omega_\tau \to M_{\tau^+}$ (conditional on (e_τ, M_{τ^-})) plus data-processing inequality yields $\kappa_{\text{processing}} \leq 1$ by construction. (H$_\kappa$) is a derived consequence, not a commitment, for Class 1.

Class 2 (Partial) and Class 3 (Coupled) require (H_κ) as a named structural commitment. Two clean sufficient conditions and a parrot-architecture worst-case witness ($\kappa_{\text{processing}} = \infty$ when $M_{\tau^+} := G$ on goals orthogonal to the latent world-state) are documented in Appendix B.3. The unconditional theorem still bounds parrot-architecture displacement directly by transferred information; the $\kappa \times \mathcal{A}$ factorization just doesn't apply.

The realized $\kappa_{processing}$ is empirically accessible. A behavioral two-goal probe — run the architecture on a fixed event set under two goal states, measure divergence of the epistemic content of the response — gives a lower bound on $\kappa_{\text{processing}}$ at the operating point. For architectures with explicit attention or routing, activation-level mediation analyses give a tighter handle. Both are out of scope for the present theory paper.

5 Mechanism

Both tracks bridge to the bound through the same move: a chain rule of relative entropy applied directly to the post-update law, marginalized over the goal. This *is* the structural composition of the two literatures — the Bayesian-inverse-problems posterior-stability machinery and the architectural classification meet at this identity.

Lemma 5.1 (Chain rule on the post-update law)**.** *Under (H1),* $\mathbb{E}_G\big[\mathrm{KL}(P_{M_{\tau^+}|e,M_{\tau^-},G} \,\|\, P_{M_{\tau^+}|e,M_{\tau^-}})\big] \;=\; I(G;\, M_{\tau^+} \mid e_\tau, M_{\tau^-})$.

The identity is exact. The right-hand side is *transferred* goal-information — the actually-realized coupling between G and the post-update model state. The left-hand side averages KL between goal-conditional and goal-marginal post-update laws. (H1)'s standard-Borel regularity is what makes this identity hold in the abstract-spaces form (Polyanskiy and Wu [20], Theorem 3.4; Gray [21], Theorem 5.4). Combined with either a transport inequality on the post-update law or a Fisher-Rao second-order expansion of KL, the bound follows by taking expectation over G and applying Jensen.

5.1 Track 1 — transport-inequality cascade

Two steps.

Step 1 Apply Lemma 5.1: $\mathbb{E}_G[\mathrm{KL}_g] = I_M$ where KL_g is the goal-conditional-vs-marginal KL at $G = g$ and $I_M = I(G; M_{\tau^+} \mid e_\tau, M_{\tau^-})$.

Step 2 Apply (H2′) slice-wise: $W_2^2(P_{M_{\tau^+}|G=g}, P_{M_{\tau^+}}) \le C_{T_2} \cdot \mathrm{KL}_g$. Take expectation over G and substitute Step 1:

$$\mathbb{E}\big[W_2^2(P_{M_{\tau^+}|G},\, P_{M_{\tau^+}})\big] \;\le\; C_{T_2} \cdot I_M.$$

Jensen on $\sqrt{\cdot}$ delivers the distance form $\mathbb{E}\, W_2 \le \sqrt{C_{T_2}\, I_M}$.

The cascade is composed from textbook information-theoretic and transport-inequality machinery — Otto-Villani for the LSI-implies-T_2 direction, Bakry-Émery for the K-strong-log-concavity-implies-LSI direction, Gozlan's characterization of T_2 via dimension-free sub-Gaussian Lipschitz concentration, Bobkov-Götze for the strictly weaker single-Lipschitz-function T_1 version. The contribution is the application of these machinery elements to the goal-conditional bias quantity on the post-update law: there is no separate posterior-pushforward step, because the chain rule lives on the post-update law directly. Full proof and (H2′) sufficient-condition verifications in Appendix B.

5.2 Track 2 — Fisher-Rao expansion

One step.

Lemma 5.2 (KL-to-Fisher-Rao second-order expansion)**.** *On a statistical manifold with Fisher information metric* **I***, for nearby P, Q:* $\mathrm{KL}(P \,\|\, Q) = \frac{1}{2} d_{FR}^2(P, Q) + O(d_{FR}^3)$.

This is the infinitesimal form of the Bregman divergence on the exponential-family Fenchel geometry — exact at second order, tight in the small-information regime (Cover and Thomas [25] §12.5; Amari and Nagaoka [18] §3.7 Theorem 3.1). Apply Lemma 5.2 slice-wise at each $G = g$ under (H4′) (every goal-conditional slice uniformly close to the goal-marginal on the manifold, so the expansion is sharp slice-wise with controlled remainder). Take expectation over G and substitute Lemma 5.1:

$$\mathbb{E}\big[d_{FR}^2(P_{M_{\tau+}|G},\, P_{M_{\tau+}})\big] \;\leq\; 2 \cdot I_M \cdot (1 + o(1)).$$

Jensen on $\sqrt{\cdot}$: $\mathbb{E}\, d_{FR} \leq \sqrt{2}\sqrt{I_M}\,(1 + o(1))$.

The $(1 + o(1))$ is the third-order remainder, controlled uniformly under (H4′) by the Amari-Chentsov tensor and the minimum Fisher eigenvalue along the goal-induced geodesic. The constant $\sqrt{2}$ is what (PI)+(R)+(K) buys: Čencov uniqueness (Theorem 4.5) forces Fisher-Rao with that exact constant.

Outside (H4′) When rare goals carry disproportionately large transferred information, or when the operating distribution sits at moderate-to-large I_M, the slice-wise expansion is no longer uniformly sharp. Track 2 has globally-valid companion bounds — Fisher-Rao at $\pi/\sqrt{2}$ via the unit L^2-sphere chord-arc bound, and a direct Hellinger statement at $1/\sqrt{2}$ — under (H1)+(PI) only, no (R), (K), or (H4′). The full taxonomy and proofs are in Appendix D. The $\pi/2 \approx 1.57$ overhead is the worst-case arc-chord ratio at the unit L^2-sphere's antipode — a geometric maximum, not a slack.

5.3 No-go — chart rescaling

The proof of Theorem 4.4 is short. Under any chart ϕ on $\mathcal{M}$ and rescaling $\phi_a := a\phi$ (a smooth coordinate change the underlying geometry treats as identity):

- *Chart-Euclidean W_2 scales linearly:* $W_2^{\phi_a}(\mu,\nu) = a \cdot W_2^{\phi}(\mu,\nu)$ — the pushforward under $x \mapsto ax$ rescales transport cost by a.
- *KL, conditional MI, Fisher-Rao geodesic, Hellinger — all chart-invariant:* relative entropies depend only on densities and dominating measures (reparameterization-invariant); Fisher metric tensor transforms covariantly so integrated geodesic distance is invariant; Hellinger as f-divergence is reparameterization-invariant.

For any architecture realizing finite positive $W_2^{\phi} > 0$ and finite positive transferred information I, the candidate universal-constant inequality would have to satisfy $a \cdot \|\Delta M_{\text{bias}}\|_{\text{Eucl},\phi} \leq C_0\sqrt{I}$ for all $a > 0$ at fixed $C_0\sqrt{I}$. Taking $a \to \infty$ contradicts the fixed right-hand side. Both required quantities ($W_2^{\phi} > 0$ and $I > 0$) are automatic on any non-trivial goal-coupled architecture — for instance, the conjugate-Gaussian Class 1 setup with any $\sigma, \tau > 0$ and a non-degenerate continuous goal distribution. The Gaussian scale family with two charts (Chart A: σ; Chart B: $\log\sigma$) is the canonical illustration: the same intrinsic geometry, Chart-A Euclidean displacement grows linearly with the operating scale while Chart-B stays bounded.

The implication is constructive. To get a universal constant the bound must be stated in a coordinate-invariant norm — one that respects the underlying geometry of the statistical manifold rather than an arbitrary chart on it. This is exactly the role Fisher-Rao plays under (PI)+(R)+(K). Alternative chart-independent commitments (TV with Pinsker, Hellinger-as-divergence, χ^2, Rényi) yield strictly weaker bounds — the no-go does not just say "commit to something intrinsic"; it says *exactly* what to commit to.

6 Conclusion

The opening question — *how much can an LLM be pulled away from what its evidence supports by the goal it is asked to satisfy?* — is the question the existing hallucination-theory lineage doesn't engage. The lineage bounds *frequency*; the question is about *size*. The two axes are orthogonal. A coupled-architecture model in (PI)-compatible operating regime can simultaneously have an inevitable-frequency hallucination floor (training-data structure, calibration constraints, learnability bottlenecks, computability obstructions) *and* a bounded-magnitude goal-coupling displacement on each event (geometric structure). The two together give a more complete picture than either alone.

What the geometric route delivers — and what the architecture-engaged commitment buys. The chain-rule move on the post-update law composes two literatures that hadn't met. Track 2 under (PI)+(R)+(K) gives a universal, dimension-free $\sqrt{2}$ constant by Čencov uniqueness; the no-go on Euclidean chart norms makes that commitment load-bearing rather than coincidental. The architectural classification reads the bound operationally: $\kappa_{\text{processing}} \times \mathcal{A}$ — coupling strength times residual ambiguity — with Class 1 (Separated) automatic by data-processing, Class 2 (Partial) and Class 3 (Coupled) named structural commitments. Plain decoder-only transformer attention is structurally Class 3 by Lemma 3.5, not as an incidental property of training.

Limitations The bounds are conditional theorems. Regularity hypotheses (H2′) and (H4′) restrict the regime each Track applies in; (PI) is an axiomatic commitment, not a derivation; the architectural factorization requires (H_κ) at corollary level. The paper is theory-only — we do not instrument a deployed LLM's belief-update dynamics. The mitigation we offer is structural: Track 1's cascade has decades of empirical use in Bayesian inverse problems; Track 2 under (PI)+(R)+(K) is mathematically airtight via Čencov; the no-go is counterexample-grade. The bound captures *one channel* of hallucination — goal-coupling-induced displacement — which is real and structurally derived. Other channels (training-data structure, calibration, capacity) live in the orthogonal frequency lineage and are addressed there.

Future directions Empirical estimation of $\kappa_{\text{processing}}$ for deployed LLMs via two-goal probing or activation-level mediation analysis; extension of the locally-tight Fisher-Rao bound to the moderate-to-large I regime via compact-manifold or staged-composition arguments; instantiation of the bound for specific architecture families (mixture-of-experts, retrieval-augmented, state-space LLMs that fall outside Lemma 3.5's attention-based scope); and connection to the frequency-lower-bound lineage of Kalai-Vempala-style results to give a layered hallucination theory in which structural-bias bounds at the per-event level compose with frequency bounds at the corpus level.

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A Failed routes

We document two derivation routes that failed at structural levels, so future work does not re-attempt them without new evidence.

A.1 Route F1: Cramér-Rao inversion

Attempt Treat G as a parameter to be estimated from Ω given (e, M). The Cramér-Rao lower bound gives $\mathrm{Var}(\hat{G}) \geq \mathbf{I}_G^{-1}$ where $\mathbf{I}_G$ is the Fisher information about G given (Ω, e, M). Invert this to a posterior-sensitivity factor: an upper bound $\|\Delta M_{\text{bias}}\| \leq L_M \cdot \sqrt{\mathbf{I}_G^{-1}}$ via a Lipschitz-in-G constant on the coupled-update mechanism.

Why it fails Cramér-Rao bounds estimator variance from *below*. To produce an *upper bound* on $\|\Delta M_{\text{bias}}\|$ via Cramér-Rao inversion would require fixing an estimator class within which $\mathbf{I}_G^{-1}$ is also an upper bound on variance — but no such estimator class exists for the coupled architecture's update mechanism. The mechanism is a deterministic function of (G, e, M), not an estimator of G, and there is no uniform-class assumption available. The direction of the Cramér-Rao bound and the direction of the bias bound are opposite; inverting one to the other requires an additional assumption that is not available.

Lesson Information-theoretic *lower-bound* machinery for estimator performance does not invert to upper bounds on side-information-induced displacement.

A.2 Route F2: Rate-distortion inversion

Attempt Apply Shannon's rate-distortion inequality $R(D) \geq h(X) - \frac{1}{2}\log(2\pi e D)$ for a Gaussian source. Invert to $D \leq \sigma^2 \cdot 2^{-2(R-h(X))}$ with "bits in" interpreted as $I(G; \Omega|e, M)$, "distortion" interpreted as $\|\Delta M_{\text{bias}}\|^2$.

Why it fails Rate-distortion theory describes the minimum bits-per-symbol required to *represent* a source within distortion D; it does not describe the maximum *displacement induced* by injecting side-information into an update mechanism. The two problems have different structures: source-coding is about optimal representations; side-information injection is about perturbation propagation. Identifying "bits in" with I and "distortion" with displacement does not produce a sound inequality — the rate-distortion curve is about a different operational quantity.

Lesson Source-coding theorems are about optimal representations of a source, not spatial displacement induced by side-information injection. Upper bounds on displacement require transport-inequality machinery (Pinsker, Otto-Villani, Bakry-Émery, Lipschitz-posterior).

A.3 The pattern

Both failed routes attempt to invert *information-theoretic lower-bound* machinery (Cramér-Rao for estimator error; rate-distortion for source coding) into *upper bounds on displacement*. The structural mismatch is uniform: lower-bound machinery for optimization quantities does not invert to upper bounds on perturbation quantities without additional structure that is not available in coupled-architecture settings. Transport-inequality machinery is on the other side of this divide and is the correct family for upper-bound-on-displacement results.

B Hypothesis verification details

This appendix collects the regularity-condition verifications referenced in Section 5.1 and Section 5.2.

B.1 (H2′) Talagrand T_2 inequality on the post-update law — sufficient conditions

(H2′) requires $W_2^2(P, P_{M_{\tau+}|e,M_{\tau-}}) \le C_{T_2}\mathrm{KL}(P \,\|\, P_{M_{\tau+}|e,M_{\tau-}})$ for all probability measures P. Three clean sufficient conditions:

Route (a): LSI on the post-update law If $P_{M_{\tau+}|e,M_{\tau-}}$ satisfies a logarithmic Sobolev inequality with constant $\rho_{\text{post}} > 0$,

$$\mathrm{Ent}_{P_{M_{\tau+}|e,M_{\tau-}}}(f^2) \;\le\; \tfrac{2}{\rho_{\text{post}}}\,\mathbb{E}_{P_{M_{\tau+}|e,M_{\tau-}}}[\|\nabla f\|^2] \quad \text{for all sufficiently smooth } f,$$

then by [26] Theorem 1, (H2′) holds with $C_{T_2} = 2/\rho_{\text{post}}$. A sufficient condition for LSI, by Bakry-Émery's curvature-dimension criterion [27], is K-strong log-concavity of the post-update density: $-\nabla^2 \log p_{M_{\tau+}|e,M_{\tau-}} \succeq K \cdot I$ for some $K > 0$, giving $\rho_{\text{post}} \ge K$.

Hypothesis (S) — own proof of strong log-concavity for Bayesian post-updates When the post-update law is itself a Bayesian posterior with negative log-likelihood $\Phi(\theta;\omega)$ and prior density $\pi(\theta)$, the posterior negative log-density is $U(\theta;\omega) = \Phi(\theta;\omega) - \log \pi(\theta)$. Strong log-concavity of the posterior is a *direct* statement about the Hessian-in-θ of U: $\nabla_\theta^2 U(\theta;\omega) \succeq K_{\text{eff}} \cdot I$ uniformly in $\omega \in \Omega$, for some $K_{\text{eff}} > 0$. Two clean sufficient conditions:

- *(S-i) Convex likelihood + strongly log-concave prior.* If $\nabla_\theta^2 \Phi(\theta;\omega) \succeq 0$ uniformly in (θ,ω) (likelihood log-concave in θ) and π is K-strongly log-concave (so $-\nabla_\theta^2 \log \pi \succeq K \cdot I$), then $\nabla_\theta^2 U \succeq K \cdot I$, giving $K_{\text{eff}} \ge K$.
- *(S-ii) Bounded negative likelihood curvature dominated by prior curvature.* If $\nabla_\theta^2 \Phi(\theta;\omega) \succeq -K_0 \cdot I$ uniformly with $K_0 < K$ (likelihood may be locally non-convex but its negative-curvature is bounded), and π is K-strongly log-concave, then $\nabla_\theta^2 U \succeq (K - K_0) \cdot I$, giving $K_{\text{eff}} \ge K - K_0 > 0$.

Either condition gives Bakry-Émery $\rho_{\text{post}} \ge K_{\text{eff}}$ and Otto-Villani $C_{T_2} \le 2/K_{\text{eff}}$. Sprungk 2020 [7] derives sharper local-Lipschitz constants for the posterior across TV / Hellinger / W_2 / KL; Cvetković and Lie 2025 [8] establish matched lower bounds; Dolera and Mainini 2023 [9] give global W_2-Lipschitz constants in terms of Fisher-information functionals — which can sharpen C_{T_2} beyond the bare $2/K_{\text{eff}}$. Mixed-Hessian smoothness $\|\partial^2_{\theta\omega}\Phi\| \le H$ is *not* a substitute: it controls how the posterior mode shifts with ω, but the LSI / strong-convexity claim is about the curvature in θ and must be made directly.

Route (b) *(corrected — bounded support gives T_1, not T_2).* Bare bounded support of diameter D on $\mathcal{M}$ is **not** a T_2-sufficient condition. *Counterexample.* On $\{0,1\} \subset \mathbb{R}$ (diameter $D = 1$), let $\nu = \frac{1}{2}\delta_0 + \frac{1}{2}\delta_1$ and $\mu_\epsilon = (\frac{1}{2} - \epsilon)\delta_0 + (\frac{1}{2} + \epsilon)\delta_1$ for $\epsilon \in (0, 1/2)$. Then $W_2^2(\mu_\epsilon, \nu) = \epsilon$ (the optimal coupling moves mass ϵ from 0 to 1), while $\mathrm{KL}(\mu_\epsilon \,\|\, \nu) = 2\epsilon^2 + O(\epsilon^4)$, so $W_2^2/\mathrm{KL} = 1/(2\epsilon) + O(1) \to \infty$ as

$\epsilon \to 0^+$ — no constant C uniformizes the ratio. The structural reason: Gozlan 2009 [22] established that T_2 is *equivalent* to dimension-free Gaussian concentration of $\mu^{\otimes n}$, strictly stronger than diameter (Talagrand-type product concentration requires LSI / curvature input, not just bounded support).

What does follow from diameter is the *weaker* T_1 inequality: by $W_1 \le D \cdot \mathrm{TV}$ and Pinsker $\mathrm{TV} \le \sqrt{\mathrm{KL}/2}$,

$$W_1(P, P_{M_{\tau^+}|e,M_{\tau^-}}) \;\le\; \tfrac{D}{\sqrt{2}}\sqrt{\mathrm{KL}(P \,\|\, P_{M_{\tau^+}|e,M_{\tau^-}})}, \qquad C_{T_1} = D^2/4,$$

sharp on the two-point Bernoulli at $p = 1/2$. (Equivalently, this is Bobkov and Götze [28] with the Hoeffding sub-Gaussian variance proxy $\sigma^2 = D^2/4$ for distributions on a diameter-D set, applied to the T_1-equivalent Lipschitz-sub-Gaussian characterization.) Substituting W_1 for W_2 in Step 2 of the Section 5.1 cascade preserves the *distance-form* bound $\mathbb{E}\|\Delta M_{\text{bias}}\| \le (D/\sqrt{2})\sqrt{I(G; M_{\tau^+}|e, M_{\tau^-})}$ at the T_1 level — at the cost of giving up Theorem 4.1's tighter squared-distance / linear-in-I refinement (5). Token-distribution belief states for LLMs on the simplex are more naturally treated under Track 2 (Fisher-Rao, Section 5.2), where the simplex's constant positive sectional curvature (on the round-sphere image $x \mapsto (\sqrt{x_1}, \ldots, \sqrt{x_n})$) provides Bakry-Émery $\mathrm{CD}(K, n-1)$ with $K > 0$, an LSI w.r.t. Fisher-Rao volume, and the universal $\sqrt{2}\sqrt{I}$ form holds without Track 1's diameter-only weakening.

Route (c): dimension-free sub-Gaussian Lipschitz concentration Per Gozlan 2009 [22], μ satisfies $T_2(C)$ iff $\mu^{\otimes n}$ enjoys dimension-free Gaussian concentration of Lipschitz functionals: there exist constants $C', \lambda > 0$ such that $\mathbb{E}_{\mu^{\otimes n}}[\exp(\lambda(f - \mathbb{E}f)^2)] \le 2$ uniformly in n for every 1-Lipschitz $f : \mathbb{R}^n \to \mathbb{R}$. Bobkov and Götze 1999 [28] characterizes the strictly weaker *single-Lipschitz-function* version (giving T_1); the Gozlan strengthening to T_2 is what makes single-distribution sub-Gaussian moments insufficient on their own. When dimension-free product concentration holds, C_{T_2} admits an explicit form in the variance proxy.

When both routes fail Heavy-tailed post-updates without LSI and without dimension-free sub-Gaussian concentration are outside Track 1's T_2 scope. The T_1 fallback under bounded support (Route (b) as corrected above) recovers the distance form at weaker constants — adequate for $\mathbb{E}\|\Delta M_{\text{bias}}\| \le C\sqrt{I}$. Track 2 under (PI) + (R) + (K) does not require (H2′) and remains available; Theorem D.2's Hellinger backstop is a global universal-constant fallback under (PI) only.

B.2 KL-to-Fisher second-order expansion

For nearby distributions $P_\theta, P_{\theta+\delta}$ on a parametric family with Fisher information $\mathbf{I}(\theta)$:

$$\mathrm{KL}(P_{\theta+\delta} \,\|\, P_\theta) = \tfrac{1}{2}\delta^\top \mathbf{I}(\theta)\delta + O(\|\delta\|^3) = \tfrac{1}{2}d_{FR}^2(P_\theta, P_{\theta+\delta}) + O(d_{FR}^3).$$

Standard derivation: Cover and Thomas [25] §12.5; Amari and Nagaoka [18] §3.7 Theorem 3.1. The $O(d_{FR}^3)$ remainder is sharp in the sense that the second-order term is exact and the third-order term is generically non-zero (off-exponential-family).

B.3 ($\mathbf{H}_\kappa$) sufficient conditions

The architectural-corollary commitment of Appendix B.3 is that $\kappa_{\text{processing}} = I(G; M_{\tau^+}|e, M_{\tau^-})/I(G; \Omega_\tau|e, M_{\tau^-})$ is bounded by some named architectural constant $\kappa^* < \infty$ (typically 1). For Class 1 (Separated) architectures, the data-processing inequality applied to the Markov chain $G \to \Omega_\tau \to M_{\tau^+}$ (conditional on (e_τ, M_{τ^-}) by architectural separation) gives $\kappa_{\text{processing}} \le 1$ automatically. For Class 2 (Partial) and Class 3 (Coupled), two structural conditions each give $\kappa_{\text{processing}} \le 1$ (or $\le 1 + \alpha$ in the second condition):

1. *Goal-blind effective kernel.* Suppose the post-update map factors as $M_{\tau^+} = K(M_{\tau^-}, e_\tau, \widetilde{\Omega}_\tau)$ with K deterministic, where $\widetilde{\Omega}_\tau$ is a goal-conditional reweighting of Ω_τ that is *non-amplifying*: $I(G; \widetilde{\Omega}_\tau|e_\tau, M_{\tau^-}) \le I(G; \Omega_\tau|e_\tau, M_{\tau^-})$. Then conditional on (e_τ, M_{τ^-}), $G \to \widetilde{\Omega}_\tau \to M_{\tau^+}$ is a Markov chain (since K depends on G only through $\widetilde{\Omega}_\tau$), and DPI applied along this chain gives

$$I(G; M_{\tau^+}|e_\tau, M_{\tau^-}) \;\le\; I(G; \widetilde{\Omega}_\tau|e_\tau, M_{\tau^-}) \;\le\; I(G; \Omega_\tau|e_\tau, M_{\tau^-}),$$

i.e., $\kappa_{\text{processing}} \le 1$.

1. *Bounded direct architectural channel.* Suppose the architecture's *direct* goal-coupling pathway (independent of Ω_τ) has bounded conditional MI relative to the resolvable channel: $I(G; M_{\tau^+}|e_\tau, M_{\tau^-}, \Omega_\tau) \;\le\; \alpha\, I(G; \Omega_\tau|e_\tau, M_{\tau^-})$ for some $\alpha \ge 0$. The chain rule of mutual information conditioned on (e_τ, M_{τ^-}) gives $I(G; M_{\tau^+}, \Omega_\tau|e_\tau, M_{\tau^-}) = I(G; \Omega_\tau|e_\tau, M_{\tau^-}) + I(G; M_{\tau^+}|e_\tau, M_{\tau^-}, \Omega_\tau)$, and using $I(G; M_{\tau^+}|e_\tau, M_{\tau^-}) \le I(G; M_{\tau^+}, \Omega_\tau|e_\tau, M_{\tau^-})$ (additional side information cannot decrease MI), we get $\kappa_{\text{processing}} \le 1 + \alpha$. The case $\alpha = 0$ — architectural channel adds nothing beyond what Ω_τ provides given (e_τ, M_{τ^-}) — gives $\kappa_{\text{processing}} \le 1$.

Worst-case witness — parrot architecture Consider a degenerate Coupled-class architecture whose update rule is $M_{\tau^+} := G$ (the post-update model state is identically the goal). Two computations:

Numerator. Since $M_{\tau^+} = G$ deterministically conditional on (e_τ, M_{τ^-}), $I(G; M_{\tau^+}|e_\tau, M_{\tau^-}) = H(G|e_\tau, M_{\tau^-})$ exactly.

Denominator. Suppose $G \perp\!\!\!\perp \Omega_\tau$ given (e_τ, M_{τ^-}) — the goal carries no information about the latent world-state beyond what the evidence and prior have already pinned down. (Operationally this is the "make me feel good" goal of Section 3.2: the user's affective goal is independent of the empirical question the evidence resolves.) Then $I(G; \Omega_\tau|e_\tau, M_{\tau^-}) = 0$.

Hence $\kappa_{\text{processing}} = H(G|e_\tau, M_{\tau^-})/0 = +\infty$. ($\text{H}_\kappa$) explicitly fails. This shows the architectural commitment is *real* for non-Separated architectures, not a free move. The unconditional theorems (Theorem 4.1, Theorem 4.1, Theorem D.2) still bound the parrot architecture's displacement by transferred information directly — the prefactor is $\sqrt{H(G|e_\tau, M_{\tau^-})}$ rather than $\sqrt{\kappa\, I(G; \Omega)}$ — but the $\kappa \times \mathcal{A}$ factorization (and hence the architectural-corollary form) does not apply.

C Conjugate-Gaussian numerical comparison

We compute the three tracks' constants explicitly for the conjugate-Gaussian observation-prior pair and exhibit the regimes in which they are tight. The setup is the one introduced in Appendix C: latent $\theta \sim \mathcal{N}(0, \tau^2)$, observation channel $\Omega|\theta \sim \mathcal{N}(\theta, \sigma^2)$, goal-conditional likelihood-mean shift $\Omega|G \sim \mathcal{N}(\beta(G), \sigma^2)$. The architecture's update is goal-blind (Class 1 Separated, conjugate Bayesian update). Identifying M_{τ^+} with the posterior-mean parameter $\mu_+ := L_{\text{post}}\Omega$ (variance $\sigma^2_{\text{post}} = L_{\text{post}}\sigma^2$ fixed), the post-update parameter law is $\mu_+|G \sim \mathcal{N}(L_{\text{post}}\beta(G),\, L^2_{\text{post}}\sigma^2)$.

Two distinct variances. The *inside-posterior covariance* $\sigma^2_{\text{post}} = L_{\text{post}}\sigma^2$ is the spread of the posterior measure on θ at a fixed observation. The *parameter-law variance* $L^2_{\text{post}}\sigma^2 = L_{\text{post}}\sigma^2_{\text{post}}$ is the spread of the random model state μ_+ under data noise. They differ by exactly L_{post}. Track 1's C_{T_2} is determined by the parameter-law variance, not the inside-posterior covariance; conflating them is the algebraic error this appendix corrects.

Transferred information and $\kappa_{\text{processing}}$. The goal-conditional law of Ω has small-$\text{Var}_G(\beta)$ MI $I(G; \Omega|e, M_{\tau^-}) \approx \text{Var}_G(\beta)/(2\sigma^2)$. The post-update parameter law inherits the same MI: the deterministic linear map $\Omega \mapsto L_{\text{post}}\Omega$ is invertible, so the data-processing inequality on the chain $G \to \Omega \to M_{\tau^+}$ saturates and $I(G; M_{\tau^+}|e, M_{\tau^-}) = I(G; \Omega|e, M_{\tau^-})$ exactly. The realized attenuation ratio is $\kappa^{\text{realized}}_{\text{processing}} = 1$. *This Class 1 conjugate-Gaussian setup does not exhibit architectural attenuation* — it sits at the worst-case Class 1 corner where DPI is tight; non-trivial $\kappa < 1$ requires a non-invertible data channel (e.g., quantization, dimension reduction) or a Class 2/3 architecture.

Track 1 (Theorem 4.1). Route (a) — LSI on the post-update parameter law $\mathcal{N}(0, L^2_{\text{post}}\sigma^2)$ at small $\text{Var}_G(\beta)$ — gives $\rho_{\text{post}} = 1/(L^2_{\text{post}}\sigma^2)$ and

$$C_{T_2} \;=\; 2/\rho_{\text{post}} \;=\; 2\, L^2_{\text{post}}\sigma^2 \;=\; \frac{2\,\tau^4\sigma^2}{(\sigma^2+\tau^2)^2}.$$

Theorem 4.1: $\mathbb{E}[W_2^2] \le C_{T_2} \cdot I(G; M_{\tau^+}|e, M_{\tau^-}) = 2L_{\text{post}}^2\sigma^2 \cdot I(G;\Omega|e, M_{\tau^-})$ (using $\kappa = 1$). Equivalently, with $\rho_{\text{LSI}} := 1/\sigma^2$,

$$\mathbb{E}[W_2^2] \le \frac{2\,L_{\text{post}}^2}{\rho_{\text{LSI}}} \cdot I(G;\Omega|e, M_{\tau^-})$$

— the canonical Stuart-school cascade constant, recovered as C_{T_2} alone (DPI saturation $\kappa = 1$ is the witness that this Class 1 example sits at the Stuart corner). The cascade is sharp: the slice-wise W_2^2/KL ratio equals C_{T_2} exactly for these Gaussian parameter laws. Limit checks: prior-dominant ($\tau \ll \sigma$): $C_{T_2} \approx 2\tau^4/\sigma^2 \to 0$; likelihood-dominant ($\sigma \ll \tau$): $C_{T_2} \approx 2\sigma^2 \to 0$; balanced ($\tau = \sigma = 1$): $C_{T_2} = 1/2$ and $\mathbb{E}[W_2^2] \le I_M/2$.

Track 2 (Theorem 4.1). Fisher metric on the parameter-law manifold $\{\mathcal{N}(\mu, L_{\text{post}}^2\sigma^2) : \mu \in \mathbb{R}\}$ (laws over the random parameter μ_+, not posterior measures on θ): $\mathbf{I}(\mu) = 1/(L_{\text{post}}^2\sigma^2)$. Fisher-Rao distance $d_{FR}(\mathcal{N}(\mu_1, v), \mathcal{N}(\mu_2, v)) = |\mu_1 - \mu_2|/\sqrt{v} = |\mu_1 - \mu_2|/(L_{\text{post}}\sigma)$. Slice-wise to the marginal $\mathcal{N}(0, L_{\text{post}}^2\sigma^2)$: $d_{FR}(P_{M_{\tau^+}|G=g}, P_{M_{\tau^+}}) \approx |\beta(g)|/\sigma$ at small $\text{Var}_G(\beta)$. Theorem 4.1 (unconditional): $\mathbb{E}[d_{FR}^2] \le 2 \cdot I(G; M_{\tau^+}|e, M_{\tau^-}) = \text{Var}_G(\beta)/\sigma^2$ — sharp on this family.

Track 2 Euclidean translation (Theorem D.4). $|\Delta\mu_+|_{\text{Eucl}} = L_{\text{post}}\sigma \cdot d_{FR}$, so $\mathbb{E}\,|\Delta\mu_+|_{\text{Eucl}} \le L_{\text{post}}\sigma\sqrt{2I_M}$. The prefactor $L_{\text{post}}\sigma = \tau^2\sigma/(\sigma^2 + \tau^2)$ is bounded above by $\tau/2$ uniformly (AM-GM at $\sigma = \tau$) and *vanishes in both extreme limits*: $L_{\text{post}}\sigma \to 0$ as $\sigma \to \infty$ (very noisy observations: the architecture learns essentially nothing from Ω, posterior-mean displacement vanishes); $L_{\text{post}}\sigma \to 0$ as $\sigma \to 0$ (sharp observations: $L_{\text{post}} \to 1$ but $\sigma \to 0$). Maximum at balanced $\sigma = \tau$. The naive "$\sigma\sqrt{2I}$" reading is wrong (correct prefactor is $L_{\text{post}}\sigma$, not bare σ); equally importantly, the "$\sigma_{\text{post}}\sqrt{2I}$" reading is *also* wrong (it conflates inside-posterior covariance $\sigma_{\text{post}}^2 = L_{\text{post}}\sigma^2$ with parameter-law variance $L_{\text{post}}^2\sigma^2$).

Track 3 — Hellinger backstop (Theorem D.2). Theorem D.2: $\mathbb{E}\,\text{Hel}(P_{M_{\tau^+}|G}, P_{M_{\tau^+}}) \le (1/\sqrt{2})\sqrt{I_M}$ globally, where Hel is between two parameter laws $\mathcal{N}(\mu_1, L_{\text{post}}^2\sigma^2)$ and $\mathcal{N}(\mu_2, L_{\text{post}}^2\sigma^2)$. Hellinger is globally bounded by 1, so the bound is non-trivial when $\sqrt{I_M/2} < 1$ — i.e., $I_M < 2$ nat.

Numerical comparison at balanced scale ($\tau = \sigma = 1$, $L_{\text{post}} = 1/2$, $L_{\text{post}}\sigma = 1/2$, $L_{\text{post}}^2\sigma^2 = 1/4$, transferred info $I_M = 1$ nat for normalization):

Table 2: Conjugate-Gaussian numerical comparison at balanced scale ($\tau = \sigma = 1$, $I_M = 1$ nat).

Track	Theorem	Bound	Numerical at $I_M = 1$
Track 1	Theorem 4.1	$\mathbb{E}[W_2^2] \le C_{T_2}\, I_M = (1/2)\, I_M$	$\mathbb{E}\, W_2 \le 1/\sqrt{2}$
Track 2	Theorem 4.1	$\mathbb{E}\, d_{FR} \le \sqrt{2I_M}$	$\mathbb{E}\, d_{FR} \le \sqrt{2}$
Track 5.5 Euclidean	Theorem D.4	$\mathbb{E}\,\|\Delta\mu_+\| \le L_{\text{post}}\sigma\sqrt{2I_M} = (1/2)\sqrt{2I_M}$	$\mathbb{E}\,\|\Delta\mu_+\| \le 1/\sqrt{2}$
Hellinger	Theorem D.2	$\mathbb{E}\,\text{Hel} \le \sqrt{I_M/2}$	$\mathbb{E}\,\text{Hel} \le 1/\sqrt{2}$

Track 1 and Track 5.5 agree exactly at balanced scale ($\mathbb{E}\, W_2 = \mathbb{E}\,\|\Delta\mu_+\|_{\text{Eucl}} = 1/\sqrt{2}$ at $I_M = 1$) — they bound the same Euclidean displacement; Track 5.5 is just Track 2 with the Fisher-Rao-to-Euclidean conversion factor $L_{\text{post}}\sigma$ on the parameter chart. The Euclidean translation goes to zero in both extreme limits (vanishing prior-dominant and vanishing likelihood-dominant). At $I_M = 1$ nat the Hellinger bound saturates the global $\sqrt{1/2} \approx 0.71$; for $I_M \ll 1$ Track 1 / Track 5.5 are tighter; for $I_M \gtrsim 2$ the Hellinger backstop is the only non-vacuous bound.

D Track 2 companions and parametric Euclidean translations

The locally-tight Fisher-Rao bound (Theorem 4.1's Track 2 instantiation) is one of three Fisher-Rao-spine bounds; the other two — a globally-valid Fisher-Rao backstop and a Hellinger backstop — recover under strictly weaker hypotheses, with universal constants of their own. Together with the conjugate-Gaussian Euclidean translation (review-defusing the no-go's apparent scale-family pathology) and the exponential-family Euclidean generalization, they form the Track 2 family.

D.1 Track 2 globally-valid backstop

Outside the uniform-locality regime — when a few rare goals carry disproportionately large transferred information, or when the operating distribution sits at moderate-to-large I_M — the slice-wise expansion's $(1+o(1))$ remainder is no longer uniformly sharp. A global Fisher-Rao bound is available under strictly weaker hypotheses, at the cost of a slightly larger constant.

Theorem D.1 (Track 2 global backstop)**.** *Under (H1) and (PI) — for all values of transferred information,*

$$\mathbb{E}\, d_{FR}\big(P_{M_{\tau^+}|e,M_{\tau^-},G},\, P_{M_{\tau^+}|e,M_{\tau^-}}\big) \;\le\; (\pi/\sqrt{2}) \cdot \sqrt{I(G;\, M_{\tau^+} \mid e_\tau, M_{\tau^-})}.$$

The constant $\pi/\sqrt{2} \approx 2.22$ *is universal, dimension-free, global — no small-information condition required, and (R), (K) are not invoked.*

Proof. On the unit L^2-sphere of $\sqrt{p}$, the chord-arc identity (Lemma D.3) gives $\mathrm{Hel}(P,Q) = \sqrt{2}\sin(d_{FR}(P,Q)/4)$ exactly. Lemma D.3 shows $d_{FR}(P,Q) \le \pi\,\mathrm{Hel}(P,Q)$ globally, with equality at the antipode Hel $= 1$. By Tsybakov [29], Lemma 2.4, $2\,\mathrm{Hel}^2 \le$ KL slice-wise. Combining: $d_{FR}^2 \le \pi^2 \mathrm{Hel}^2 \le (\pi^2/2)\,\mathrm{KL}$ slice-wise. Take $\mathbb{E}_G$, substitute the chain rule Lemma 5.1: $\mathbb{E}\, d_{FR}^2 \le (\pi^2/2)\, I_M$. Jensen on $\sqrt{x}$: $\mathbb{E}\, d_{FR} \le (\pi/\sqrt{2})\sqrt{I_M}$. □ □

The $\pi/2 \approx 1.57$ overhead vs. the locally-tight $\sqrt{2}$ is exactly the worst-case arc-chord ratio at the unit L^2-sphere's antipode — a geometric maximum, not a slack. Equality is achieved in the limit Hel $\to 1$ at the antipode, when goals can drive the post-update measure to a near-disjoint support from the marginal.

D.2 Hellinger backstop

The chord companion to Theorem D.1's arc bound, with the cleanest universal constant.

Theorem D.2 (Hellinger backstop)**.** *Under (H1) and (PI) — for all values of transferred information,*

$$\mathbb{E}\,\mathrm{Hel}\big(P_{M_{\tau^+}|e,M_{\tau^-},G},\, P_{M_{\tau^+}|e,M_{\tau^-}}\big) \;\le\; (1/\sqrt{2}) \cdot \sqrt{I(G;\, M_{\tau^+} \mid e_\tau, M_{\tau^-})}.$$

Under (H_κ), the architectural factorization recovers: $\mathbb{E}\,\mathrm{Hel} \le (1/\sqrt{2})\sqrt{\kappa_{\mathit{processing}} \cdot I(G;\, \Omega \mid e, M)}$.

Proof. By Tsybakov [29], Lemma 2.4, $2\mathrm{Hel}^2(P,Q) \le \mathrm{KL}(P\|Q)$ globally for all probability measures. Apply slice-wise at each $G = g$, take expectation over G, and substitute the chain rule Lemma 5.1. □ □

Hellinger satisfies (PI) — it is an f-divergence with $f(t) = (\sqrt{t}-1)^2/2$ preserved under Markov morphisms (Csiszár 1967; Liese-Vajda 1987) — and is locally proportional to Fisher-Rao on the unit-sphere representation. Throughout we adopt the *standard statistician's* Hellinger convention $\mathrm{Hel}^2(P,Q) := \frac{1}{2}\|\sqrt{p}-\sqrt{q}\|_{L^2}^2 = 1 - \int \sqrt{pq}$, so Hel $\in [0,1]$, and the *Amari-Nagaoka* Fisher-Rao convention $d_{FR}(P,Q) = 2\arccos\int\sqrt{pq}$ (twice the great-circle arc on the unit sphere of $\sqrt{p}$ in L^2, $d_{FR} \in [0,\pi]$). Different conventions in the literature differ by factors of 2; the forms above are the ones consistent with Hel $\in [0,1]$, $d_{FR} \in [0,\pi]$, and the $2\mathrm{Hel}^2 \le$ KL inequality used in the proof.

The three Fisher-Rao-spine bounds — locally-tight $\sqrt{2}$ (under (H1)+(H4′)+(PI)+(R)+(K)), globally-valid $\pi/\sqrt{2}$ (Theorem D.1, under (H1)+(PI) only), and Hellinger chord $1/\sqrt{2}$ (Theorem D.2, under (H1)+(PI) only) — share the same intrinsic geometry and have strictly nested hypotheses. There is no "metric switch" between regimes: macroscopic hallucination is in Fisher-Rao directly via Theorem D.1; Theorem D.2 is the chord companion on the same unit L^2-sphere.

D.3 Hellinger-Fisher-Rao chord-arc geometry

Lemma D.3 (Global chord-arc bound on the unit L^2-sphere)**.** *For any probability measures P, Q, $d_{FR}(P,Q) \le \pi \cdot \mathrm{Hel}(P,Q)$, with equality at the antipode* $\mathrm{Hel}(P,Q) = 1$ *(correspondingly $d_{FR}(P,Q) = \pi$).*

Proof. From $d_{FR} = 4\arcsin(\mathrm{Hel}/\sqrt{2})$, define $\phi : (0,1] \to (0,\pi]$ by $\phi(h) := 4\arcsin(h/\sqrt{2})/h$, so $d_{FR}/\mathrm{Hel} = \phi(\mathrm{Hel})$. Set $g(h) := 4\arcsin(h/\sqrt{2})$. Differentiating: $g'(h) = 4/\sqrt{2-h^2}$ for $h \in [0,\sqrt{2})$, $g''(h) = 4h/(2-h^2)^{3/2} > 0$ on $(0,\sqrt{2})$ — so g is *strictly convex* on this interval, with $g(0) = 0$. By the secant-slope monotonicity for convex functions vanishing at zero, $\phi(h) = g(h)/h$ is monotonically increasing on $(0,\sqrt{2})$, with limits $\lim_{h\to 0^+}\phi(h) = g'(0) = 4/\sqrt{2} = 2\sqrt{2}$ and $\phi(1) = g(1) = 4\arcsin(1/\sqrt{2}) = \pi$. Hence $\phi(h) \le \pi$ for all $h \in (0,1]$, with equality at $h = 1$. □ □

Local proportionality As $d_{FR} \to 0$, Hel $\approx d_{FR}/(2\sqrt{2})$, so Theorem 4.1's Track 2 locally-tight bound $\mathbb{E}\, d_{FR} \le \sqrt{2I_M}$ jointly with Lemma D.3 gives $\mathbb{E}\,\mathrm{Hel} \lesssim (1/2)\sqrt{I_M}$ in the small-information limit — *tighter* than Theorem D.2's global $(1/\sqrt{2})\sqrt{I_M}$ by a factor of $\sqrt{2}$. So in the (H4′) regime, Theorem 4.1 + Lemma D.3 jointly give the tightest Hellinger bound; Theorem D.2 (under (PI) only, no (H4′)) is the global statement.

D.4 Conjugate-Gaussian Euclidean translation

When (H1) is realized by a parametric family with known Fisher information, the locally-tight Fisher-Rao bound translates to an explicit Euclidean bound on parameter displacement. The conjugate-Gaussian case is paradigmatic and review-defusing: *Euclidean displacement does not blow up* in the prior-dominant limit, contrary to a naive reading of the no-go.

Theorem D.4 (Conjugate-Gaussian Euclidean bound)**.** *Under the conjugate-Gaussian setup ($\theta \sim \mathcal{N}(0,\tau^2)$, $\Omega \mid \theta \sim \mathcal{N}(\theta,\sigma^2)$, goal-conditional likelihood-mean shift $\Omega \mid G \sim \mathcal{N}(\beta(G),\sigma^2)$), the post-update parameter law is the Gaussian $\mu_+ \mid G \sim \mathcal{N}(L_{post}\beta(G),\, L_{post}^2\sigma^2)$ with $L_{post} = \tau^2/(\sigma^2+\tau^2)$. The locally-tight Fisher-Rao bound translates to the Euclidean bound on random-parameter displacement*

$$\mathbb{E}\,|\Delta\mu_+|_{Eucl} \;\le\; L_{post}\,\sigma \cdot \sqrt{2I_M} \cdot (1+o(1)),$$

with prefactor $L_{post}\,\sigma = \tau^2\sigma/(\sigma^2+\tau^2) \le \tau/2$ uniformly. Under (H_κ), $\mathbb{E}\,|\Delta\mu_+|_{Eucl} \le L_{post}\sigma\sqrt{2\kappa_{processing}\, I(G;\Omega \mid e, M)}(1+o(1))$.

Proof. The post-update parameter law $\mu_+ \mid G$ is Gaussian with mean $L_{\text{post}}\beta(G)$ and variance $L_{\text{post}}^2\sigma^2$ (variance preserved across goals because σ^2_{post} is fixed by the conjugate-update structure). The Fisher metric on the manifold $\{\mathcal{N}(\mu, L_{\text{post}}^2\sigma^2) : \mu \in \mathbb{R}\}$ in the μ-coordinate is $\mathbf{I}(\mu) = 1/(L_{\text{post}}^2\sigma^2)$, so the Fisher-Rao geodesic distance between two such laws is $d_{FR}(\mu_1,\mu_2) = |\mu_1-\mu_2|/(L_{\text{post}}\sigma)$ — a flat metric pulled back from $\mathbb{R}$. Hence $|\Delta\mu_+|_{\text{Eucl}} = L_{\text{post}}\sigma \cdot d_{FR}$. Take expectations and substitute the locally-tight Fisher-Rao bound. □ □

The prefactor $L_{\text{post}}\sigma = \tau^2\sigma/(\sigma^2+\tau^2)$ is bounded above by $\tau/2$ uniformly (AM-GM at the maximizer $\sigma = \tau$) and *vanishes in both extreme limits*: $L_{\text{post}}\sigma \to 0$ as $\sigma \to \infty$ (very noisy observations: the architecture learns essentially nothing from Ω regardless of goal) and $L_{\text{post}}\sigma \to 0$ as $\sigma \to 0$ (very sharp observations: $L_{\text{post}} \to 1$ but $\sigma \to 0$, so even though the posterior tracks data tightly the data variance vanishes). Maximum at balanced $\sigma = \tau$. The naive "$\sigma\sqrt{2I}$" reading is wrong (correct prefactor is $L_{\text{post}}\sigma$); the "$\sigma_{\text{post}}\sqrt{2I}$" reading is *also* wrong (it conflates inside-posterior covariance $\sigma^2_{\text{post}} = L_{\text{post}}\sigma^2$ with parameter-law variance $L_{\text{post}}^2\sigma^2$). The Section 4.1 no-go still applies as stated to *charts*: $L_{\text{post}}\sigma$ is family-specific.

D.5 Exponential-family Euclidean generalization

For an exponential family $P_\theta(x) = h(x)\exp(\theta^\top T(x) - A(\theta))$ with natural parameter $\theta \in \Theta \subseteq \mathbb{R}^d$ and log-partition A, the Fisher information metric is $\mathbf{I}(\theta) = \nabla^2 A(\theta)$. The infinitesimal Fisher-Rao expansion gives $d_{FR}^2(P_{\theta_1},P_{\theta_2}) = (\theta_1-\theta_2)^\top \mathbf{I}(\theta^*)(\theta_1-\theta_2) + O(\|\theta_1-\theta_2\|^3)$ for θ^* on the geodesic between θ_1,θ_2. So Euclidean displacement on the natural parameter satisfies $\|\theta_1-\theta_2\|^2 \le \mathbf{I}_{\min}^{-1}\cdot d_{FR}^2$ where $\mathbf{I}_{\min}$ is the minimum eigenvalue of $\mathbf{I}$ along the geodesic.

Theorem D.5 (Exponential-family Euclidean bound)**.** *For an exponential family in natural-parameter coordinates with Fisher-information minimum-eigenvalue* $\mathbf{I}_{\min}$ *along the goal-induced geodesic, the locally-tight Fisher-Rao bound translates to*

$\mathbb{E}\,\|\Delta\theta_{bias}\|_{Eucl} \;\leq\; \mathbf{I}_{\min}^{-1/2} \cdot \sqrt{2I_M} \cdot (1 + o(1)),$

and under (H_κ): $\mathbb{E}\,\|\Delta\theta_{bias}\|_{Eucl} \leq \mathbf{I}_{\min}^{-1/2}\sqrt{2\kappa_{processing}\, I(G;\Omega \mid e, M)}(1 + o(1)).$

Proof. By Theorem 4.1's Track 2 instantiation, $\mathbb{E}\, d_{FR}^2 \leq 2I_M(1 + o(1))$. By the infinitesimal Fisher-Rao expansion, $\|\theta_1 - \theta_2\|^2 \leq \mathbf{I}_{\min}^{-1} d_{FR}^2$. Take expectations and Jensen. □ □

Worked examples *Bernoulli on logit chart, restricted to $p \in [\delta, 1-\delta]$:* $\mathbf{I}(\theta) = p(1-p)$ has no positive lower bound on the full chart $\theta \in \mathbb{R}$; restricting to $p \in [\delta, 1-\delta]$ gives $\mathbf{I}_{\min} \geq \delta(1-\delta)$ and finite, dimension-free Euclidean bound $\mathbb{E}\|\Delta\theta\|_{\text{Eucl}} \leq [\delta(1-\delta)]^{-1/2}\sqrt{2I_M}$ — contingent on boundary avoidance. *Multinomial on log-ratio chart:* Fisher minimum eigenvalue bounded below by $\min_i p_i$ on the simplex interior; for δ-bounded models, $\mathbb{E}\|\Delta\theta\|_{\text{Eucl}} \leq \delta^{-1/2}\sqrt{2I_M}$. *Gaussian-mean chart with fixed variance v:* $\mathbf{I}(\mu) = 1/v$ constant on the manifold; $\mathbb{E}\|\Delta\mu\|_{\text{Eucl}} \leq \sqrt{v}\sqrt{2I_M}$. *Gaussian-scale chart $\theta = \sigma$:* $\mathbf{I}(\sigma) = 2/\sigma^2$ — $\mathbf{I}_{\min}^{-1/2} = \sigma/\sqrt{2}$ grows with σ, recovering the no-go's pathology consistent with Section 4.1; reparameterizing to $\theta = \log\sigma$ gives $\mathbf{I}(\log\sigma) = 2$ constant, restoring a finite Euclidean bound.

The pattern: for natural-parameter charts on simplex / discrete-outcome models with the goal-induced geodesic restricted away from the boundary, the Euclidean Track 2 translation is dimension-free and finite; for scale-family parametrizations off the natural-parameter chart, the prefactor grows in σ or λ^{-1} — recovering Section 4.1's pathology consistent with the no-go.

E Proofs of main results

This appendix collects full proofs deferred from §4–§5 of the main text. Proof structure: each section restates the formal claim from main text, then provides the full derivation. Auxiliary lemmas used in proofs are stated where they first arise.

E.1 Track 1 cascade — full derivation

The umbrella theorem's Track 1 instantiation (Theorem 4.1 under (H1) + (H2′)) follows from a two-step composition: chain rule on the post-update law, then slice-wise Talagrand T_2.

Proof. Step 1 — Chain rule of relative entropy on the post-update law. By Cover and Thomas 2006 [25] Theorem 2.5.3 applied to the conditional law of M_{τ^+} given (e_τ, M_{τ^-}) marginalized over G:

$$\mathbb{E}_G\big[\mathrm{KL}(P_{M_{\tau^+}|e,M_{\tau^-},G} \,\|\, P_{M_{\tau^+}|e,M_{\tau^-}})\big] \;=\; I(G; M_{\tau^+} \mid e_\tau, M_{\tau^-}).$$

The (H1) standard-Borel regularity makes this hold in the abstract-spaces form (Polyanskiy and Wu [20], Theorem 3.4; Gray [21], Theorem 5.4). The right-hand side is *transferred* goal-information into the post-update model state.

Step 2 — Slice-wise Talagrand T_2. Apply Hypothesis 4.2 slice-wise at each $G = g$:

$$W_2^2(P_{M_{\tau^+}|e,M,G=g},\, P_{M_{\tau^+}|e,M}) \;\leq\; C_{T_2} \cdot \mathrm{KL}(P_{M_{\tau^+}|e,M,G=g} \,\|\, P_{M_{\tau^+}|e,M}).$$

Take expectation over G and substitute Step 1:

$$\mathbb{E}\big[W_2^2(P_{M_{\tau^+}|G},\, P_{M_{\tau^+}})\big] \;\leq\; C_{T_2} \cdot I(G; M_{\tau^+} \mid e_\tau, M_{\tau^-}).$$

Jensen's inequality on $\sqrt{x}$ delivers the distance form

$$\mathbb{E}\, W_2(P_{M_{\tau^+}|G},\, P_{M_{\tau^+}}) \;\leq\; \sqrt{C_{T_2}} \cdot \sqrt{I(G; M_{\tau^+} \mid e_\tau, M_{\tau^-})}. \;\square \qquad \square$$

The architectural-corollary form (Corollary 4.7 under (H_κ)) follows by substituting $I(G; M_{\tau^+} \mid e, M_{\tau^-}) \leq \kappa_{\text{processing}} \cdot I(G; \Omega_\tau \mid e, M_{\tau^-})$ on the right-hand side; the substitution is *definitional* per (4), with the boundedness of $\kappa_{\text{processing}}$ being the substantive content of (H_κ).

E.2 Track 2 cascade — full derivation

The umbrella theorem's Track 2 instantiation (under (H1) + (H4′) + (PI) + (R) + (K)) follows in one step using the KL-to-Fisher-Rao expansion.

Proof. Apply Lemma 5.2 under (H1) + (H4′) slice-wise at each $G = g$ to get

$$d_{FR}^2(P_{M_{\tau^+}|e,M,G=g},\, P_{M_{\tau^+}|e,M}) \;\le\; 2\,\mathrm{KL}(P_{M_{\tau^+}|G=g} \,\|\, P_{M_{\tau^+}})\,(1 + o(1)).$$

Under (H4′) the remainder is controlled uniformly: with bounded Amari-Chentsov tensor $|T_{ijk}\delta^i\delta^j\delta^k| \le \mathcal{T}_\star\|\delta\|^3$ on the goal-induced neighborhood and minimum Fisher eigenvalue $\mathbf{I}_{\min} > 0$, the slice-wise inequality $d_{FR}^2(P_{M_{\tau^+}|G=g}, P_{M_{\tau^+}}) \;\le\; 2\,\mathrm{KL}_g(1 + R_3(\delta_\star))$ holds with $R_3(\delta_\star) = (\mathcal{T}_\star/3)\sqrt{2\delta_\star/\mathbf{I}_{\min}^3} \to 0$ as $\delta_\star \to 0$.

Take expectation over G and substitute the chain-rule identity from Lemma 5.1:

$$\mathbb{E}\big[d_{FR}^2(P_{M_{\tau^+}|G},\, P_{M_{\tau^+}})\big] \;\le\; 2 \cdot I(G; M_{\tau^+} \mid e_\tau, M_{\tau^-}) \cdot (1 + o(1)).$$

Jensen on $\sqrt{x}$ delivers the distance form

$$\mathbb{E}\, d_{FR}(P_{M_{\tau^+}|G},\, P_{M_{\tau^+}}) \;\le\; \sqrt{2} \cdot \sqrt{I(G; M_{\tau^+} \mid e_\tau, M_{\tau^-})} \cdot (1 + o(1)). \;\square$$

□

The constant $\sqrt{2}$ emerges from the second-order coefficient $\frac{1}{2}$ in the KL-to-Fisher expansion (Amari and Nagaoka[18] §3.7 Theorem 3.1) — exactly what (K) crystallizes as a normalization. Outside (H4′), the locally-tight bound's $(1 + o(1))$ remainder is no longer uniformly sharp; companion bounds in Appendix D cover the moderate-to-large I_M regime under strictly weaker hypotheses (Track 2 global, Hellinger backstop).

E.3 No-go on Euclidean chart norms — full proof

Theorem 4.4's proof uses a chart-rescaling lemma plus a one-line consequence.

Lemma E.1 (Chart-rescaling sensitivity of Euclidean displacement)**.** *Let $\mathcal{M}$ be a statistical manifold and $\phi : U \to \mathbb{R}^d$ a chart on an open set $U \subseteq \mathcal{M}$. For any $a > 0$, write $\phi_a := a\phi$ for the rescaled chart. For every pair of probability laws μ, ν on U (in particular, the goal-conditional and goal-marginal post-update laws of* (3)*):* (a) *The chart-Euclidean Wasserstein distance scales linearly: $W_2^{\phi_a}(\mu, \nu) = a \cdot W_2^{\phi}(\mu, \nu)$.* (b) *Conditional KL, conditional mutual information, Fisher-Rao geodesic distance, and Hellinger distance are all chart-invariant: they take the same value under ϕ and ϕ_a.*

Proof. (a) The pushforward under the linear map $x \mapsto ax$ rescales Euclidean distances by a, hence rescales any optimal coupling's transport cost. *(b)* KL depends only on the underlying densities and dominating measure (Radon-Nikodym derivatives are reparameterization-invariant). The Fisher metric tensor transforms covariantly under chart change, so the integrated geodesic distance is invariant. Hellinger as an f-divergence is reparameterization-invariant. □ □

Proof. Proof of Theorem 4.4. Suppose for contradiction that a coordinate-independent $C_0 < \infty$ exists such that for every chart ϕ on $\mathcal{M}$ and every architecture satisfying (H1)+(H2′) with non-zero post-update displacement,

$$\|\Delta M_{\text{bias}}\|_{\text{Eucl},\,\phi} \le C_0\sqrt{I(G; M_{\tau^+} \mid e_\tau, M_{\tau^-})}.$$

For any architecture realizing $W_2^{\phi}(P_{M_{\tau^+}|G,e,M_{\tau^-}}, P_{M_{\tau^+}|e,M_{\tau^-}}) > 0$ and finite positive transferred information, applying the supposed bound under the rescaled chart ϕ_a gives, by Lemma E.1,

$$a \cdot \|\Delta M_{\text{bias}}\|_{\text{Eucl},\,\phi} \;=\; \|\Delta M_{\text{bias}}\|_{\text{Eucl},\,\phi_a} \;\le\; C_0\sqrt{I},$$

with the right-hand side unchanged across charts. Taking $a \to \infty$ contradicts the fixed $C_0\sqrt{I}$. The same conclusion holds for the architectural-corollary form $C\sqrt{\kappa_{\text{processing}} \cdot I(G; \Omega \mid e, M)}$ under (H_κ): both sides inherit Lemma E.1's chart-(non)invariance, and the contradiction goes through verbatim.
□ □

The proof avoids type-shifts between Dirac points and pushforward laws, never invokes (H2′)-violating two-point constructions, and does not assert any specific value of I — it only uses that some architecture realizes finite positive W_2^ϕ and finite positive I, automatic on any non-trivial goal-coupled architecture (e.g., the conjugate-Gaussian setup with any $\sigma, \tau > 0$ and a non-degenerate continuous goal distribution). The Gaussian scale family with two charts (Chart A: σ; Chart B: $\log \sigma$) is the canonical illustration: same intrinsic geometry, Chart-A Euclidean displacement grows linearly while Chart-B stays bounded as the operating scale grows.

E.4 Čencov uniqueness and sharpness — full proof

Theorem 4.5 has two parts.

Proof. Proof of (a) — Uniqueness of the metric. (PI) + (R) invokes Čencov's uniqueness theorem [17] (modern treatment Ay et al. [30], Theorem 5.1): on $\mathcal{M}$, the Fisher information metric $\mathbf{I}$ is the unique (up to a global positive scalar $c > 0$) Riemannian metric invariant under Markov morphisms. Write $d_{\mathcal{M}} = \sqrt{c}\, d_{FR}$ where d_{FR} is at the Amari-Nagaoka normalization $\mathbf{I}(\theta) = -\mathbb{E}[\nabla^2 \log p_\theta]$. The KL-to-Fisher second-order expansion (Amari and Nagaoka [18] §3.7 Theorem 3.1) gives $\mathrm{KL}(P_\theta \| P_{\theta+\delta}) = \frac{1}{2} d_{FR}^2 + O(d_{FR}^3)$, so $\frac{1}{2} d_{\mathcal{M}}^2 = (c/2) d_{FR}^2 + O(d_{FR}^3)$. Constraint (K) requires $\frac{1}{2} d_{\mathcal{M}}^2$ agree with KL at second order, forcing $c = 1$ and $d_{\mathcal{M}} = d_{FR}$. □

Proof. Proof of (b) — Sharpness of the constant. The conjugate-Gaussian Class 1 (Separated) family of Appendix C with $\mathrm{Var}_G(\beta) \to 0$ and $I_M = \mathrm{Var}_G(\beta)/(2\sigma^2)$ held fixed has $\mathbb{E}[d_{FR}^2(P_{M_{\tau^+}|G}, P_{M_{\tau^+}})] = \mathrm{Var}_G(\beta)/\sigma^2 = 2I_M$ exactly (verified in Appendix C). Hence $\sup\{\mathbb{E}\, d_{FR}^2 / I_M$: instances satisfying (H1) + (PI) + (R) + (K) + (H4′)$\} \geq 2$, so any candidate constant $C^2 < 2$ fails on this family. Combined with (a), $C = \sqrt{2}$ is the unique sharp upper-bound constant. □ □

The upshot: once we adopt (PI) at full Markov-morphism strength, plus Riemannian structure, plus KL as the cascade's information coordinate, the *only* such-invariant choice of norm on $\mathcal{M}$ is Fisher-Rao, and the *unique sharp* upper-bound constant is $\sqrt{2}$. Alternative chart-independent commitments — TV with Pinsker, Hellinger-as-divergence, χ^2, Rényi — give weaker bounds: TV-Pinsker is non-tight (the constant is universal but the inequality is loose except at coincident measures); Hellinger-as-divergence and χ^2-as-divergence violate (R); switching divergences on the RHS violates (K). The global Theorem D.1's $\pi/\sqrt{2}$ is also sharp in its own (PI)-only regime — achieved in the limit $\mathrm{Hel} \to 1$ at the unit L^2-sphere's antipode.

E.5 Coupled-class attention connectivity — full proof

Proof. Proof of Lemma 3.5. By induction on layer depth.

Layer 0 (embedding). $h_0^{(i)}$ depends only on X_i — the position-i token embedding. Goal positions i_G produce embeddings $h_0^{(i_G)}$ that are functions only of the goal tokens at those positions; same for evidence positions i_E. No cross-position dependence yet.

Layer $\ell \geq 1$. A standard transformer block (pre-LN or post-LN) computes

$$h_\ell^{(j)} = h_{\ell-1}^{(j)} + \mathrm{Attn}_\ell(\mathrm{LN}(h_{\ell-1}))^{(j)} + \mathrm{MLP}_\ell(\mathrm{LN}(h_{\ell-1}^{(j)})),$$

where LayerNorm and the MLP are *position-wise* (the operation at position i depends only on $h_{\ell-1}^{(i)}$, not on other positions). This induces a directed position-graph with two edge types at each layer: (i) a *same-position* residual edge $h_{\ell-1}^{(j)} \to h_\ell^{(j)}$ always present (via the residual stream); (ii) *cross-position* attention edges $h_{\ell-1}^{(i)} \to h_\ell^{(j)}$ present whenever $\mathrm{Attn}_\ell^{(j,i)}(\theta) \neq 0$. Position-wise LayerNorm and MLP preserve both edge types.

Inductive claim. For every $\ell \geq 0$ and every input position $i \leq j$, $h_\ell^{(j)}$ has a directed-graph path back to position i in the input.

Inductive step. The same-position residual chain gives $i = j$. For $i \neq j$ with $i \leq j$, non-degenerate attention at any single layer $\ell' \leq \ell$ along the path gives a cross-position edge $h_{\ell'-1}^{(i)} \to h_{\ell'}^{(j)}$, which composes with the residual chain on either side to yield a directed-graph path from input position i to $h_\ell^{(j)}$.

Coupled-class conclusion. For any goal position $i_G \leq j$, $h_\ell^{(j)}$ has a path back to position i_G in the input whenever attention is non-degenerate on the (j, i_G) edge at some layer $\ell' \leq \ell$. So under non-degenerate attention, G (encoded at positions i_G) is causally upstream of every quantity contributing to a post-update model state read off from position $j \geq \max(i_G \cup i_E)$. Class 3 (Coupled) by construction.

Robustness to architectural variants. - RMSNorm, GroupNorm, FlashAttention preserve the position-graph: position-wise normalization in the first two cases; mathematically-equivalent attention implementation in the third. - Causal masking preserves $\text{Attn}_\ell^{(j,i_G)} \neq 0$ for $i_G \leq j$ — the regime the lemma covers. - Sliding-window or sparse attention preserves connectivity by composition across layers when the window/sparsity pattern admits a multi-hop path from i_G to j, which is the case for trained LLMs in their operating regime. - Architectures with explicit goal-to-output attention masking, or token-dropping at intermediate layers, fall outside the lemma's scope and are appropriately Class 2 (Partial). □ □

Lemma 3.5 establishes a *downstream-output graph-reachability* claim: at every layer and every position causally downstream of the goal and evidence under standard causal masking, the activation has a directed-graph path back to at least one goal position. *Three caveats:*

- *Graph reachability is not the same as causal-influence magnitude.* A directed-graph path with near-zero attention weights produces a formal edge with negligible quantitative influence on $h_\ell^{(j)}$. The connectivity claim is therefore *structural* — about the existence of a path, not the magnitude of effect — and translates into Class 3 (Coupled) classification under the Markov-blanket conditional-independence reading. Bounding $\kappa_{\text{processing}}$ requires the architectural-bandwidth conditions of Appendix B.3 (information-bottleneck on the goal channel; bounded direct-channel capacity uniformly over operating distributions).
- *Causal masking restricts the lemma's scope to downstream positions.* For $j < \min(i_G)$ — positions strictly upstream of any goal token — there is no causal path from goal to $h_\ell^{(j)}$ under causal attention masks; those positions are Class 1 (Separated) in their own right.
- *Sparse and sliding-window attention require a multi-hop reachability assumption.* The connectivity proof composes single-layer attention edges across layers; for sliding-window or sparse-attention architectures, multi-hop connectivity from i_G to j requires at least one composition of single-layer paths through the window/sparsity pattern. Architectures whose sparsity pattern excludes goal-to-output communication entirely fall outside the lemma's hypothesis and are Class 2 (Partial).

The bias-bound *interpretation* — bounding the displacement of an implicit posterior under goal-conditional re-prompting — additionally invokes the in-context-learning correspondence [12–15] between the next-token distribution and an implicit posterior. The structural Coupled-class claim of Lemma 3.5 is robust without it; the bias-bound *application* is conditional on the in-context-learning correspondence, which has its own scope (in-distribution prompts: tight; OOD/jailbreak: degraded to a bound on output-distribution displacement rather than belief-state displacement).

F Extended related work

§2 Related Work names the two main lineages this paper composes (LLM hallucination theory; Bayesian inverse-problems posterior stability). This appendix covers the additional adjacent strands — Fisher-Rao Bayesian sensitivity, Owhadi-Scovel-Sullivan brittleness, mean-field variational inference stability, and the Markov-blanket apparatus from active inference — that sit close enough to require explicit cite-and-distinguish but do not resolve the goal-coupling-displacement gap.

F.1 Strand 1: LLM hallucination theory

The recent statistical-inevitability and capacity-impossibility lineage targets the *frequency* of hallucination on architecture-agnostic grounds. Kalai and Vempala 2023 [1] establish that calibrated language models must hallucinate at a rate close to the fraction of singleton facts in the training data — a statistical lower bound that holds independently of the transformer architecture. Kalai et al. 2026 [2] extend this to argue that hallucinations originate as binary-classification errors compounded by training-and-evaluation incentives that reward guessing under uncertainty. Karbasi et al. 2025 [3] reduce hallucination *detection* to language identification in the Gold-Angluin framework, showing that detection is impossible from positive examples alone but feasible with expert-labeled negatives. Wu et al. 2024 [4] establish a VC-dimension-based no-free-lunch result for non-hallucinating learning. Suzuki et al. 2025 [5] give a probabilistic positive result — hallucinations can be made statistically negligible — counterbalancing the computability-theoretic inevitability arguments. Guo and Li 2026 [31] frame hallucination as a rate-distortion theorem for membership testing under capacity constraints. Liu et al. 2025 [32] formalize hallucinations as estimation failures with high-probability lower bounds on the hallucinate rate. Chlon et al. 2025 [33] frame transformers as Bayesian-in-expectation-not-realization with $O(\log n)$ permutation dispersion. Zeng and al 2026 [34] decompose hallucination risk into data-driven and reasoning-driven components.

Our distance from this strand: we bound a *displacement*, not a frequency, under named *geometric* assumptions, and we engage the *architecture's coupling structure* directly via $\kappa_{\text{processing}}$. The bound is not a substitute for the frequency bounds; it is an orthogonal axis. Among the recent papers, Zeng et al.'s decomposition is closest in shape to our $\kappa \times I$ factorization but differs on structural axis (training-time vs inference-time, vs our architectural-vs-informational).

F.2 Strand 2: Bayesian inverse-problems posterior stability

The transport-inequality and Lipschitz-posterior machinery driving Track 1 is mature. Stuart 2010 [6] establishes well-posedness of Bayesian inverse problems on function spaces, with Lipschitz continuity of the posterior in the data. Sprungk 2020 [7] extends this to local Lipschitz stability with respect to perturbations of the prior and the log-likelihood, in TV / Hellinger / Wasserstein / KL distances. Cvetković and Lie 2025 [8] match the upper bounds with lower bounds, emphasizing that posterior sensitivity *increases* as the posterior concentrates. Dolera and Mainini 2023 [9] obtain Lipschitz-continuity of probability kernels in the optimal-transport framework, with explicit constants in terms of Fisher-information functionals and weighted Poincaré constants — the closest-in-spirit prior result to Track 1's form. Garbuno-Iñigo et al. 2023 [10] develop integral-probability-metric perturbation analysis under log-Sobolev-style functional inequalities. Latz 2020 [35] provides well-posedness under weaker continuity. [? lie-sullivan-teckentrup-2017] give Hellinger-distance bounds for randomized forward models. Hosseini et al. 2025 [11] develop conditional optimal transport on function spaces for amortized Bayesian inference, with explicit regularity estimates on conditioning maps from prior to posterior — the closest single-paper match for Track 1's cascade. Dolera et al. 2024 [36] derive Wasserstein-based posterior contraction rates. Del Moral 2026 [37] gives Sinkhorn / Schrödinger bridge contraction via log-Sobolev — most recent transport-inequality-plus-LSI work, adjacent to Track 1's machinery.

Our distance from this strand: we apply this machinery to *belief-goal-coupled* inference under an architectural classification. None of these papers address goal-conditional bias under explicit architectural-coupling factorization. The closest, Hosseini-Hsu-Taghvaei, is amortized Bayesian inference rather than goal-coupling bias; the cascade overlaps but the application differs.

F.3 Strand 3: Fisher-Rao geometry for Bayesian sensitivity

Kurtek and Bharath 2015 [38] provide a Fisher-Rao framework for Bayesian sensitivity analysis under ε-contamination classes — the closest existing Fisher-Rao deployment to ours. Same geometric machinery; different target (sensitivity vs universal-constant bias bound). Lebanon 2004 [39] extends Čencov-Campbell uniqueness to conditional information geometry — load-bearing precursor for the conditional-distribution setting in which our bound lives. Čencov 1982 [17] is the original uniqueness result for the Fisher metric under invariance under sufficient statistics; Amari and Nagaoka 2000 [18] give the modern treatment of the KL-to-Fisher second-order expansion. McCulloch 1989 [40] and Ruggeri and Wasserman 1993 [41] develop earlier Fisher-information-flavored sensitivity analyses; both are foundational but produce neither a universal constant nor a goal-coupling bound.

F.4 Strand 4: Bayesian brittleness (handle carefully)

Owhadi et al., Owhadi et al. 2015, 2015 [23,24] establish that Bayesian inference is brittle under finite information in continuous settings: arbitrarily close models on the same data can yield arbitrarily different posterior conclusions, and any prior + model can be perturbed slightly to achieve any desired posterior. Owhadi and Scovel 2017 [42] develops the qualitative-robustness companion analysis.

A reader might worry that the brittleness results contradict our Track 1 stability bound. They do not. Owhadi-Scovel-Sullivan's no-go is for *arbitrary stability under finite information* in fixed metrics (TV, generalized moments) — they construct adversarial perturbations *outside* the well-posed regime. Our Theorem 4.1 (Track 1 instantiation) carries explicit (H1) + (H2$'$) hypotheses that exclude their brittleness regime: (H2$'$)'s post-update concentration is precisely the regularity that brittleness requires *failing*. The two no-gos constrain different things; our Theorem 4.4 is for *universal-constant absent the (PI) + (R) + (K) triple*, which is a third, distinct statement. The honest reading: three different no-go results, three different scope conditions, none implying or contradicting the others.

F.5 Strand 5: Adjacent recent work

Sheng et al. 2025 [43] establish stability of mean-field variational inference under strongly log-concave targets — adjacent to Track 1's transport-inequality machinery. Cattiaux and Guillin 2021 [44] develop functional inequalities (Poincaré, Cheeger, log-Sobolev) for perturbed measures with explicit constants — background machinery for Hypothesis 4.2. Ley et al. 2015 [45] give Wasserstein-distance bounds for nested densities via the Stein method. Cvetković et al. 2023 [46] address observation-operator choice for misspecified inverse problems. Zahm et al. 2018 [47] provide certified dimension reduction for nonlinear Bayesian inverse problems. Kleijn and van der Vaart 2006 [48] study misspecified Bayesian inference asymptotics; Shalizi 2009 [49] studies dynamics of Bayesian updating under dependent data and misspecified models. Seo et al. 2026 [50] derive W_2-stability bounds for tangent-linearized Gaussian inference on smooth manifolds — adjacent for the manifold-Bayesian-inference framing.

In LLM-stability-flavored work: Xu 2025 [51] (*Policy Cliff*) analyzes RLHF stability for LLMs/LRMs through reward-policy maps; Su-Kempe-Ullrich 2024 address jailbreaking from a statistical perspective; Biau and Boyer 2026 [52] develop statistical proxies for retrieval-augmented generation; Hyeon et al. 2026 [53] address action-sufficient goal representations in RL. None of these target the goal-coupling bias quantity (3) under an architectural-coupling factorization.

F.6 Strand 6: Active inference and Markov-blanket apparatus

The Markov-blanket apparatus from active inference [54–56] supplies the conditional-independence machinery our architectural-separation classification invokes. We adopt the Pearl-blanket reading [16] — the technical conditional-independence statement — and explicitly do not adopt the Friston-blanket metaphysical reading. The Goal/Update Coupling Class partition (Class 1 Separated / Class 2 Partial / Class 3 Coupled), with the explicit Coupled-class scope-exit, is the structural extension that the Pearl-blanket form invites but does not by itself produce.

F.7 Relation to existing decompositions

Zeng and al 2026 [34]'s HalluGuard framework decomposes hallucination risk into *data-driven* and *reasoning-driven* components, linked respectively to training-time mismatches and inference-time instabilities. This is the closest-shape decomposition to our $\kappa_{\text{processing}} \times I$ factorization. The structural axes differ: HalluGuard's axis is *temporal* (training-time vs inference-time errors); ours is *architectural-vs-informational* (where the goal can flow vs how much it could resolve). The two decompositions live at different layers and do not reduce to each other. A model can have low data-driven hallucination (clean training) and still exhibit large goal-coupling displacement (Coupled-class architecture in a high-ambiguity regime), and vice versa.

Chlon et al. 2025 [33] frame transformers as *Bayesian in expectation but not in realization*, with permutation-induced order effects scaling as $O(\log n)$. This is structurally distinct from our framing — we treat the architecture as performing a single goal-conditioned belief update on a fixed input, not as averaging over orderings — but the two are compatible: their Quantified Martingale Violation bounds the deviation from Bayesian-in-realization behavior, and our bound captures one specific channel (goal-coupling) of how non-Bayesian-in-realization a coupled architecture can be.

F.8 Failed routes — pointer

Two derivation routes failed at structural levels and are documented in Appendix A: (F1) Cramér-Rao inversion fails because Cramér-Rao bounds estimator variance from below, and inverting it for an upper bound on side-information-induced displacement requires a fixed-estimator-class assumption not available in coupled-architecture settings. (F2) Rate-distortion inversion fails because rate-distortion theory describes the minimum bits-per-symbol required to *represent* a source within distortion D, not the maximum *displacement induced* by injecting side-information into an update. The pattern: information-theoretic *lower-bound* machinery for estimator or encoder performance does not invert to upper bounds on side-information-induced displacement.